

# Identity Security Cloud (ISC) Platform

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## VERSION 5.0 (INDUSTRY READY)

## Handbook

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# Identity Security Cloud Platform

Identity Security Cloud simplifies identity governance. From day one, it allows you to automate and control user provisioning, access requests, separation of duties policies, and access certification for auditors.

With the power of AI-driven recommendations, you can make intelligent access choices, keeping your organization secure and compliant.

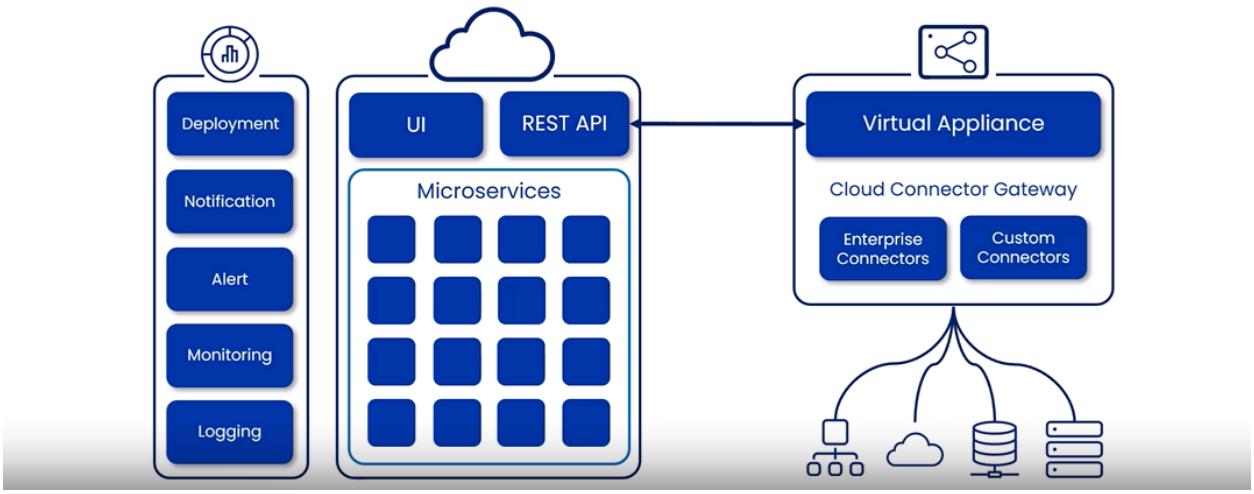
## Identity Governance Benefits

Identity governance can help your organization effectively address today's complex organizational challenges. Identity Security Cloud does this by balancing four critical objectives.

If you have taken the Discover Identity Security with SailPoint course, you are already familiar with these critical objectives. If not, you can

## Identity Security Cloud Incorporates a Multi-tenant Architecture

Identity Security Cloud is built from the ground up using microservices. As a multi-tenant SaaS architecture, Identity Security Cloud requires minimal infrastructure overhead. You can securely deploy and administer identity services from our SailPoint Identity Security Cloud platform. Here is a short video that explains this concept further.



## Identity Security Cloud Authorization Levels

It can be beneficial for organizations to hand off certain responsibilities to specific individuals securely. Sharing responsibilities through various authorization (user) levels ensures administrators have little responsibility or power over governance actions.

There are several available authorization levels in Identity Security Cloud. These levels act as security profiles, granting specific access permissions to pages and functions within Identity Security Cloud.

## User

The lowest and most common level. If a user's authorization level is undefined, this is the default enforced by Identity Security Cloud. Users have no elevated access. They can do work only related to their access, such as updating their passwords, security questions, or phone numbers. They can also respond to work assigned to them by someone else or by one of the automated processes within Identity Security Cloud.

## Organizational Administrator

Org is another name for Tenant. The Identity Security Cloud Org Admin authorization level should be granted to your system super users. The Org Admin user can visit all screens, run all commands, view all data, and update all data across your tenant. Users with this level of access can do anything within Identity Security Cloud that has been exposed to customer users. The User and the Org Admin authorization levels are the two extremes of the access scale.

Based on an individual's responsibilities, an Identity Security Cloud Org Admin can grant permission to specific product functions through specialized administration levels.

## HelpDesk

With the Help Desk authorization level, users can help other users with application account issues. However, users with help desk authorization cannot make configuration changes to sources, apps, and other features in Identity Security Cloud. Their access includes viewing details of a user's identity and current permissions.

They can also view activity related to recent logins and changes to a user's account access configuration. Their ability to view user activity and interact with identity data applies to all identities.

## Certification Administrator

Certification Administrator permission allows users to create, monitor, and manage certification campaigns.

## Report Administrator

Report Administrator permission allows users to search and save reports everywhere they are available.

## Role Administrator

Role Administrator permission allows users to create, manage, and edit roles and search and save reports from selected pages in Identity Security Cloud.

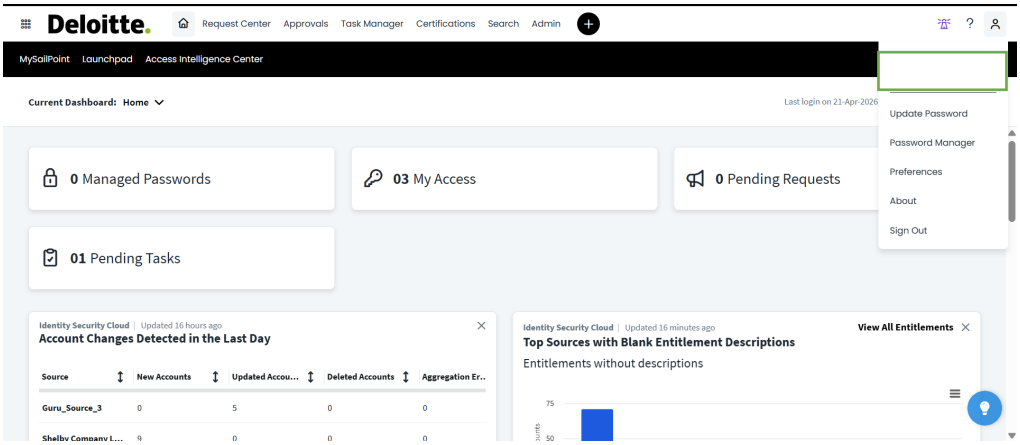
# Source Administrator

The Source Administrator permission allows users to create, configure, manage, and edit sources and access profiles.

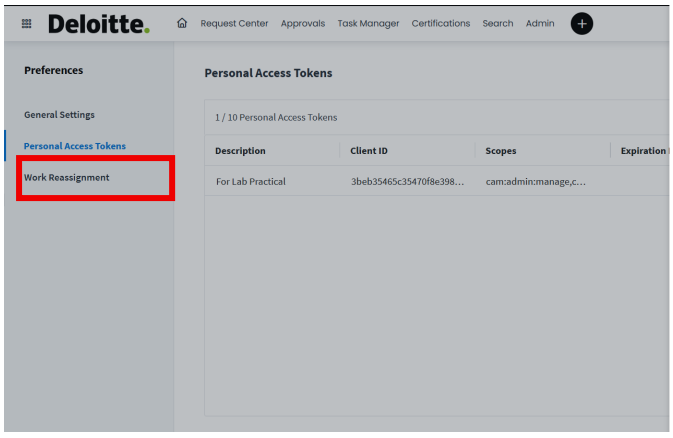
# ISC integration with VS Code:

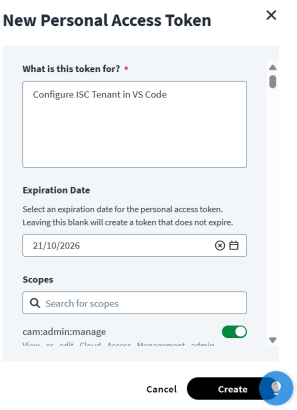
## Steps to Configure ISC Tenant in VS Code:

1. **Click on Profile → Preferences**




2. **Create New Personal Access Token**



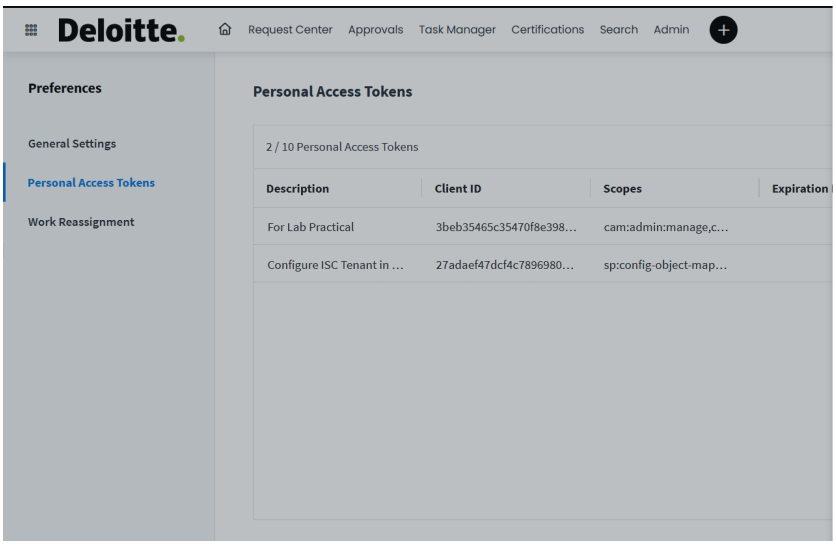



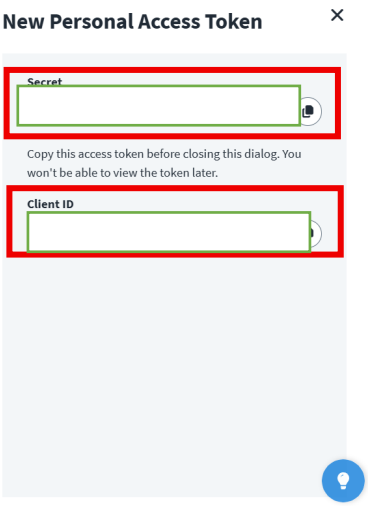

**Enter the details:**

1. What is the token used for

2. **Expiration Date**: Expiration Date of PAT
3. **Scopes**: Scopes required from PAT
4. **Save Client ID & Secret:**

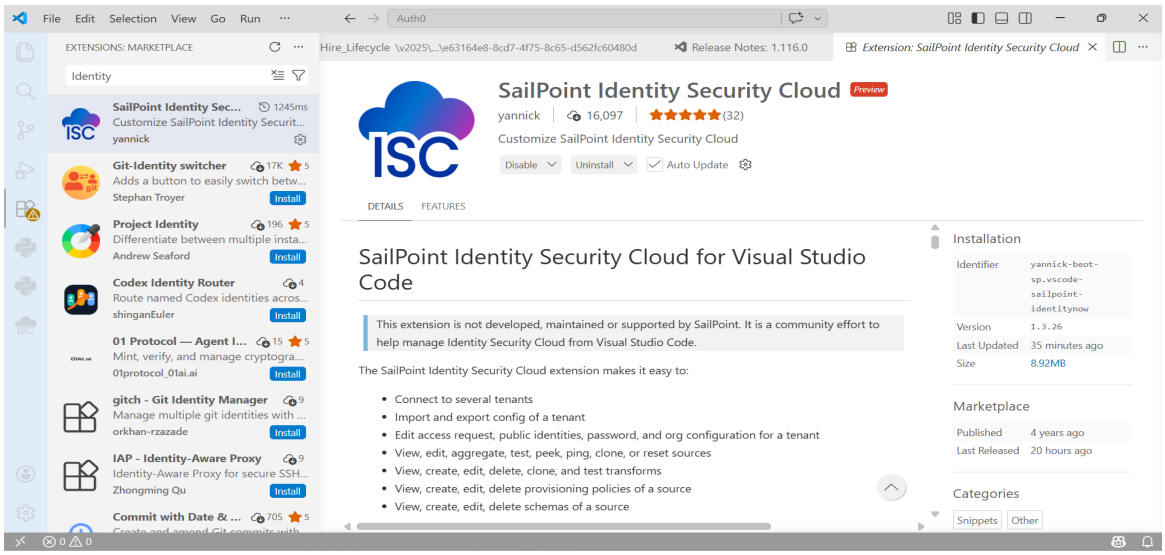
After successful creation of PAT, Secret & Client ID will be populated. Store it.



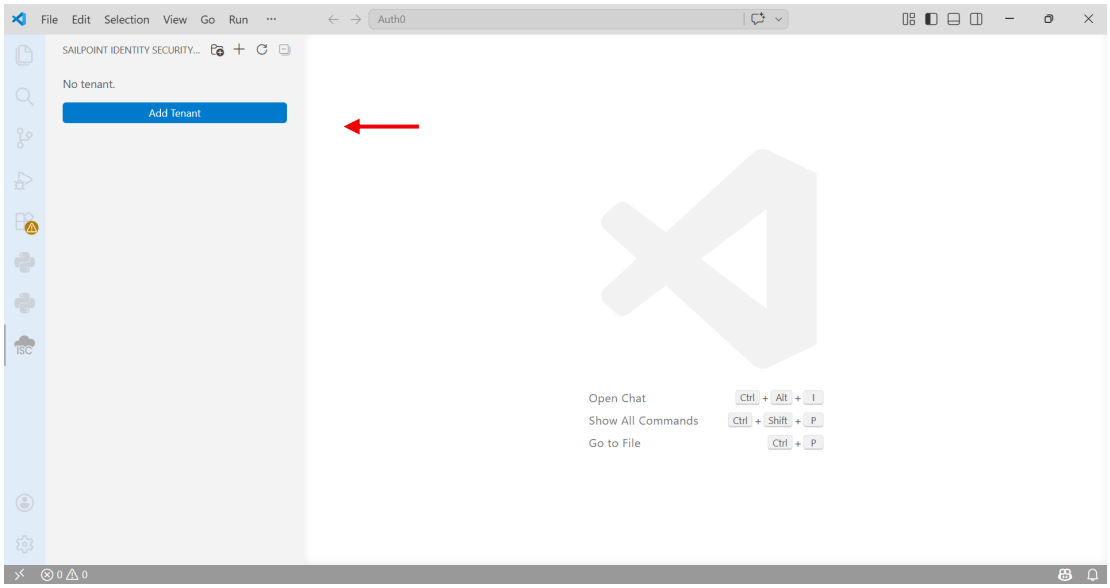

5. **Download ISC Extension in VS Code:**

**Go to VS Code → Extensions** - Download SailPoint Identity Security Cloud Extension

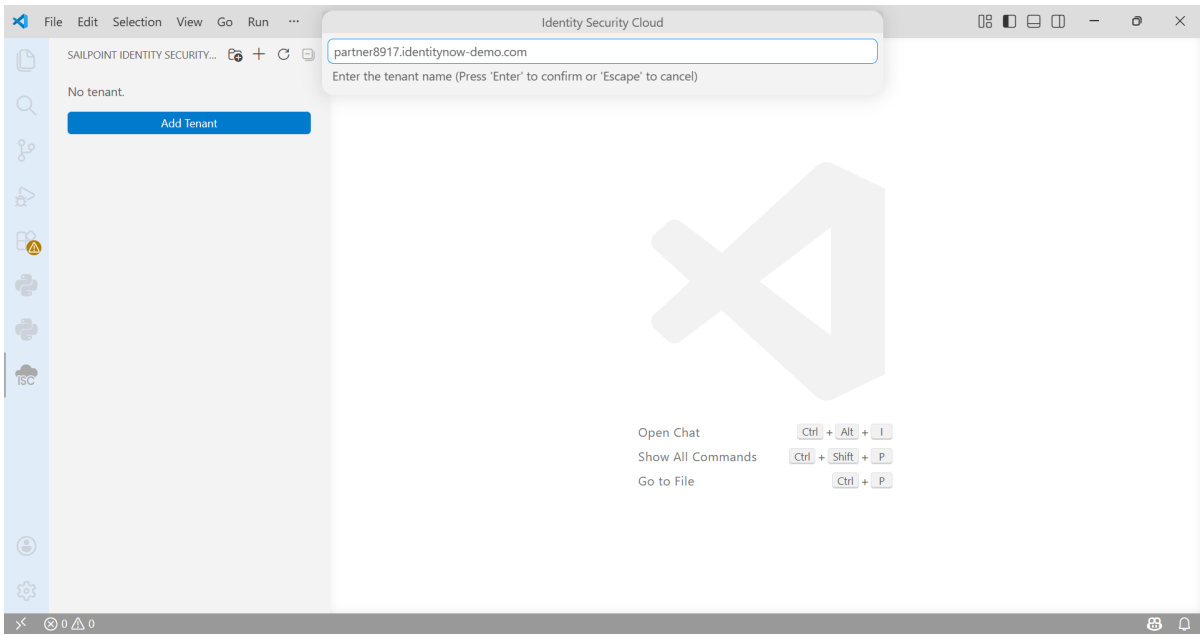


## 6. Add Tenant:

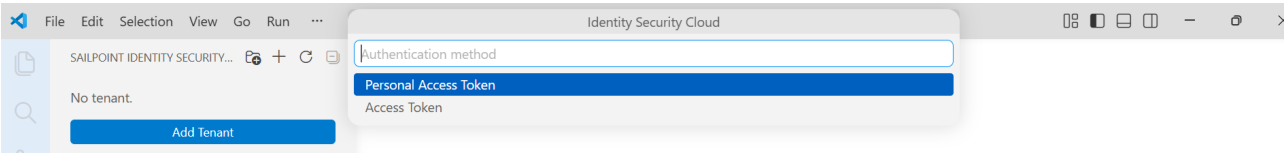
**Go to ISC tab in Side Bar → Click on Add Tenant**




## 7. Enter Tenant URL:

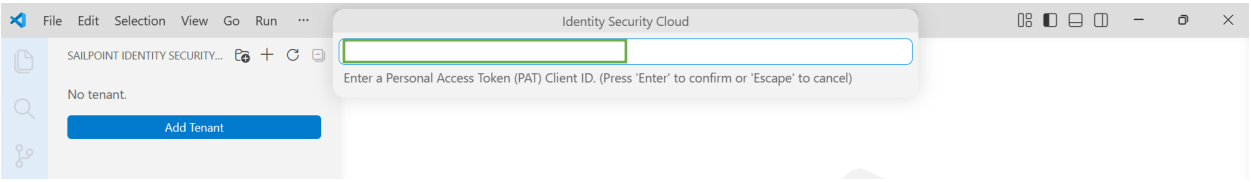


8. **Select Authentication Method:**
   **Options:**
   1. Personal Access Token
   2. Access Token

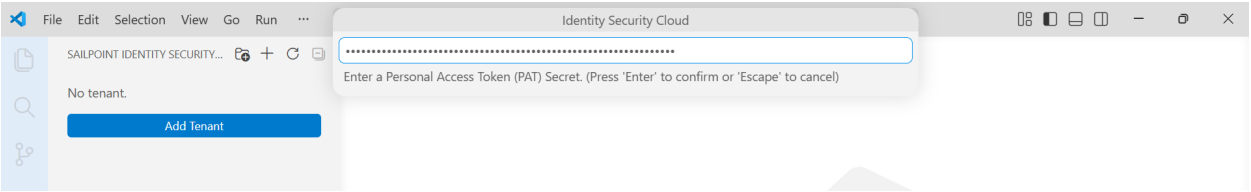



We Selected Personal Access Token.

9. **Enter Client ID:**




10. **Enter Secret:**

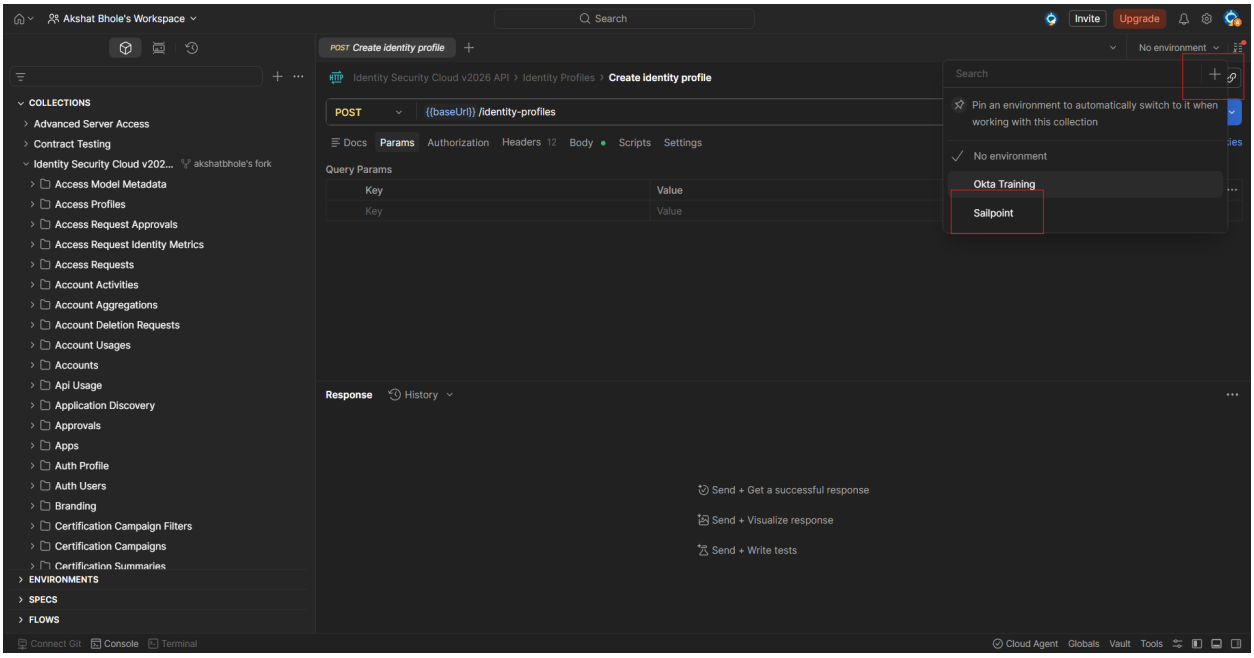



11. **ISC Tenant Configured in VS Code Successfully**

# ISC integration with Postman:

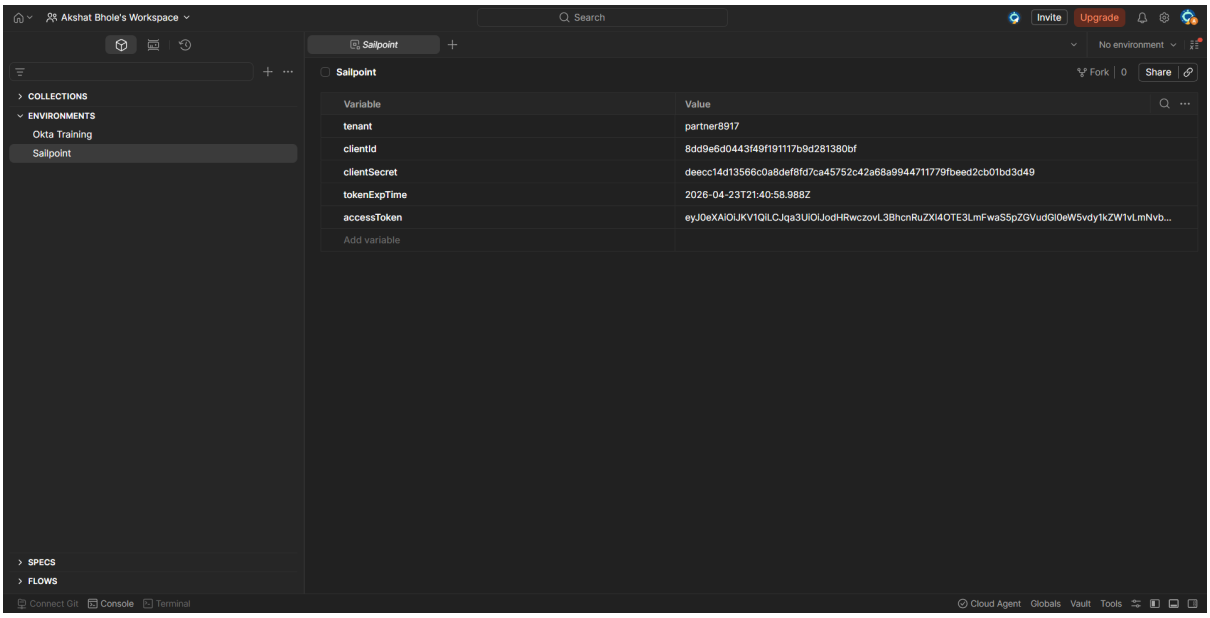
**Step 1: Create a New Environment in Postman**

- Open Postman
- Click Environments → New Environment
- Name it: SailPoint_ISC_Dev (or anything you like)




Add these variables:

| Variable Name | Initial Value |
|---|---|
| tenant | your-tenant-name |
| client_id | your-client-id |
| client_secret | your-client-secret |
| access_token | (leave empty) |




**Step 2: Generate Access Token**

Create a new request:

- Method → POST
- URL:

https://{{tenant}}.api.identitynow.com/oauth/token

Go to Body → x-www-form-urlencoded

Add:

| Key | Value |
|---|---|
| grant_type | client_credentials |
| client_id | {{client_id}} |
| client_secret | {{client_secret}} |

Click Send

**Step 3: Make Your First API Call**

Example: Get all identities

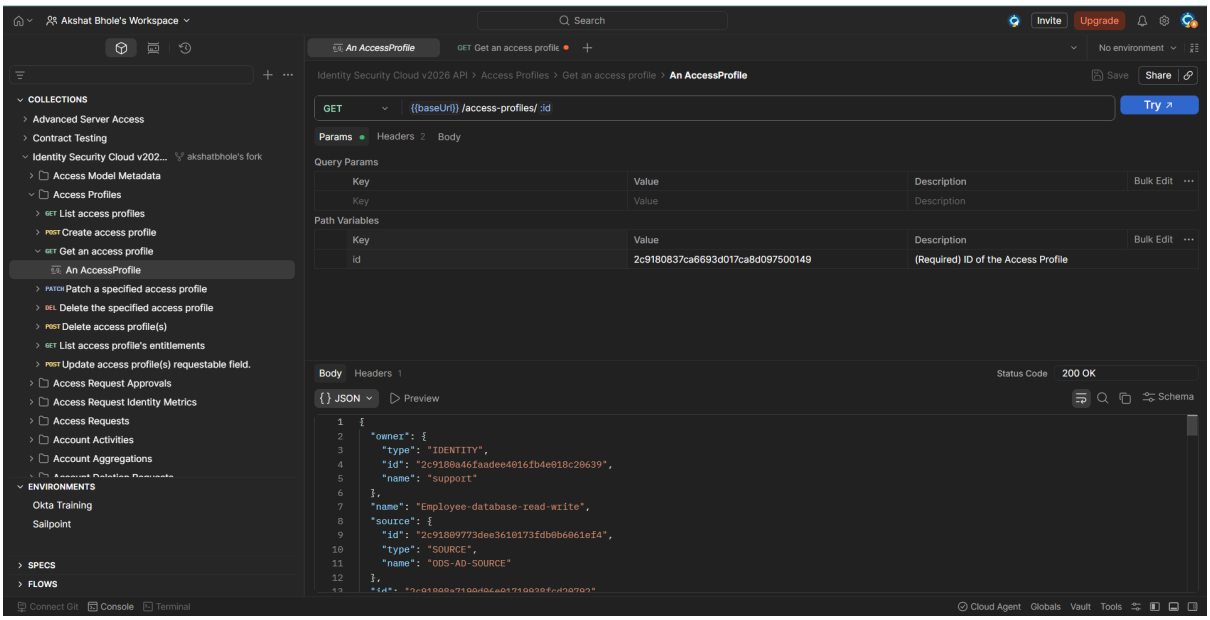
- Method → GET
- URL:

https://{{tenant}}.api.identitynow.com/v3/identities

Go to Headers:

| Key | Value |
|---|---|
| Authorization | Bearer {{access_token}} |
| Content-Type | application/json |

Click Send



## 6. Final Summary

**First create environment**



**Then generate token**



**Store token**



**Use token to call APIs**

**The Identity Model**

The very foundation of Identity Security Cloud functionality is Identity. Every Identity Security Cloud project begins by loading Identities from your data sources to build an identity model for your organization. An Identity Model comprises Identities, Accounts, and corresponding Access Privileges.

**Identity**

An Identity represents a person and the governance information associated with them, including security-relevant attributes and their current status in the organization.

**Account**

An Account is a representation of an Identity on a source system. This user account contains information about the identity and may also contain access privileges.

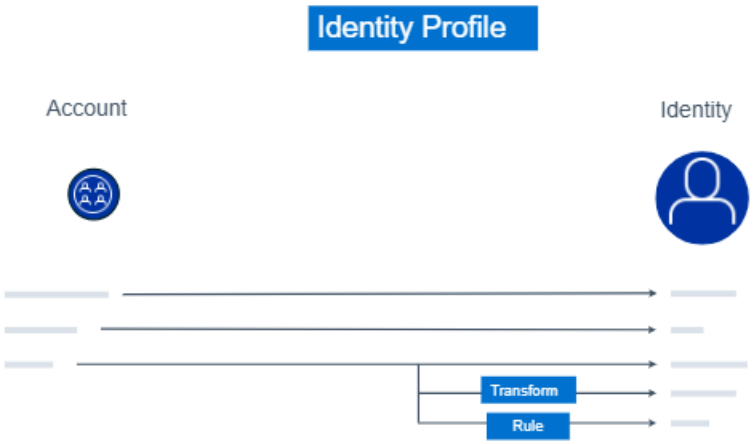
**Access**

Access defines what a user can and cannot do in your organization. Access can be entitlements, authorizations, groups, rights, or permissions. Access can also be defined through Roles and Access Profiles.

**Creating Identity Security Cloud Accounts**

Creating Identities in Identity Security Cloud begins with identifying at least one authoritative source. An Authoritative Source is any third-party application, database, or directory management system that provides a dependable list of people in your organization to whom you want to govern access. Identities can be employees, contractors, vendors, or customers.

# Identity Profile




# Data Aggregation

**Account Schema:** Employees

| Attribute Name | | Description | Type |
|---|---|---|---|
| id | Account ID | The unique ID for the account | string |
| name | Account Name | The name of the account - typical username etc | string |
| givenName | | The first or given name of the user associated with the acc… | string |
| familyName | | The last, family name, or surname of the user associated w… | string |
| manager | | Reference to the manager of this particular account | string |
| email | | | string |
| locale | | | string |
| location | | The location associated with this account | string |
| department | | | string |
| title | | | string |

Correlation is the process of associating a user account with a correlating Identity.

Identity Security Cloud performs the correlation process during the aggregation of accounts from sources.

# The Access Model

The Identity Security Cloud Access Model is built on access building blocks. These building blocks, entitlements, access profiles, roles, and lifecycle states work together to model access privileges in your organization.

## Access Model Components

**Entitlements**

Entitlements specify the type of access or permissions a user has when logging into an application and are associated with a user account. They define what a user can and cannot do with their account. Entitlements may be rights, such as read or write, or they can be security features of the native system, such as an administrator. Generally speaking, entitlements describe the lowest level of access supported on the native system.

**Access Profiles**

An access profile groups one or more entitlements from the same source and represents combinations of logically related access. Multiple entitlements from a single source can be bundled into one or more access profiles.

**Roles**

Roles are business-focused groupings of entitlements and access. They relate to the access an identity should have based on something you know about that person, such as job title, department, location, or peer group.

**Lifecycle States**

The Lifecycle state defines a person's employment status within your organization, such as when a person is onboarded, changes jobs, retires, or is released.

Each identity can reside in only one lifecycle state, but it can also have multiple roles, multiple access profiles, and multiple entitlements.

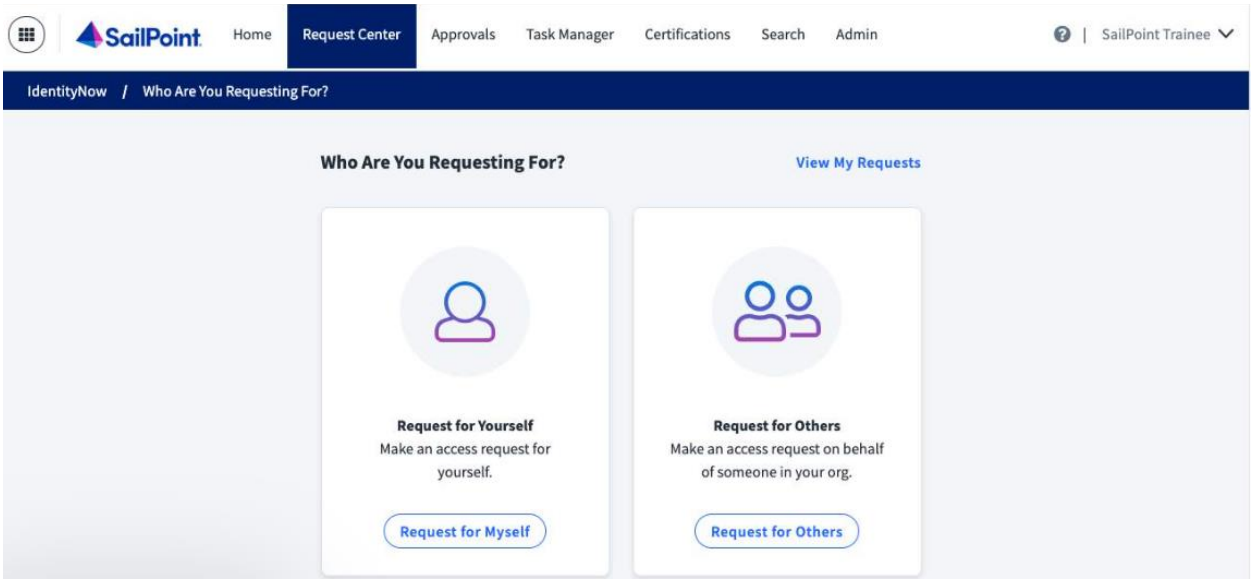
**Access Requests**

An Access Request is an interactive process of asking for a specific role, access profile, or entitlement.

**How Access Request Works**

In the Identity Security Cloud Request Center, users can submit requests for additions to their access and make requests for others in their organization. The submitted request passes through the approval gatekeepers before moving to the provisioning steps, where the access changes are fulfilled on the native systems. Identity Security Cloud allows for multiple options when making requests.

1. You may choose to allow users to make requests only for themselves.
2. Your managers can make requests for their direct reports.
3. You can allow any user to make requests for any other user.



# Access Certification

**What is Access Certification?**

At the highest level, access certification takes a user's set of assigned access and presents that data to an authority such as their manager or a source owner, who then reviews the access to verify that it is correct. The authority can approve access, revoke access, or reassign approval to another authority.

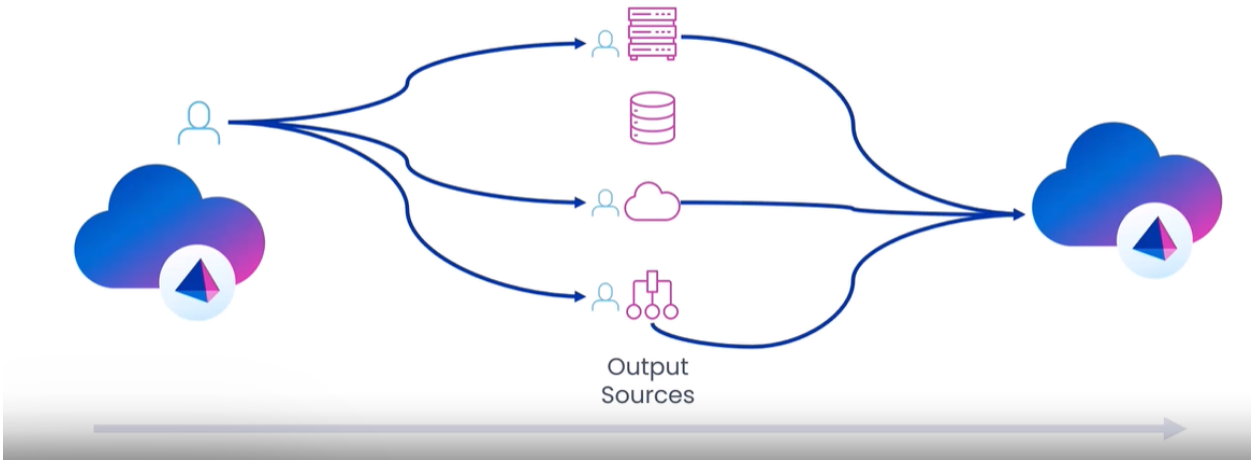
The Certifications service is an interactive review process that allows assigned users to oversee what users can access from various sources. Each round of reviews is called a Certification Campaign.

# Certification Campaign

A Certification Campaign is generally performed periodically within an organization to review user access to IT-controlled systems. Your organization will have specific reasons for performing certifications. Among the most common are legal and audit requirements that dictate that the organization validates personnel and their access to data systems.

# Provisioning

Provisioning is granting, changing, or removing user access to systems, applications, and databases based on a unique user identity.




## The Provisioning Process

The Provisioning process is about managing access. Whenever Identity Security Cloud reaches out to a connected source to make a change, it is provisioning.

You can best understand the process by considering the steps that initiate and fulfill the process.

- ***Step 1***

**Trigger Provisioning**

A triggering event, such as a user action or an automated event in Identity Security Cloud, results in a request for data change.

- ***Step 2***

**Package the Plan**

Identity Security Cloud packages the request into a plan describing all the data and the operations to communicate these changes across all systems.

- ***Step 3***

**Manage Fulfillment**

Identity Security Cloud sends the request for fulfillment directly, for connected sources, or by notification to an external ticketing system or an individual to manually fulfill the request.

- ***Step 4***

**Verify the Change**

After the request has been fulfilled, the data is re-aggregated, and the requested changes verified.

## Triggering Provisioning

Two categories of actions can trigger provisioning: Data-driven/automated and Request-driven/user-initiated.

In the Data-driven/automated category, changes in identity data can trigger provisioning per lifecycle state configuration or per role assignment logic. Sometimes, you can propagate identity changes to other systems to keep specific attributes in sync across multiple systems.

In the Request-driven/user-initiated category, the process begins with a user action or request. Users can request access, which is then sent for approval and added to their accounts on other systems. They can also submit a password change request. Even a revocation of certification results in access changes that follow this provisioning flow.

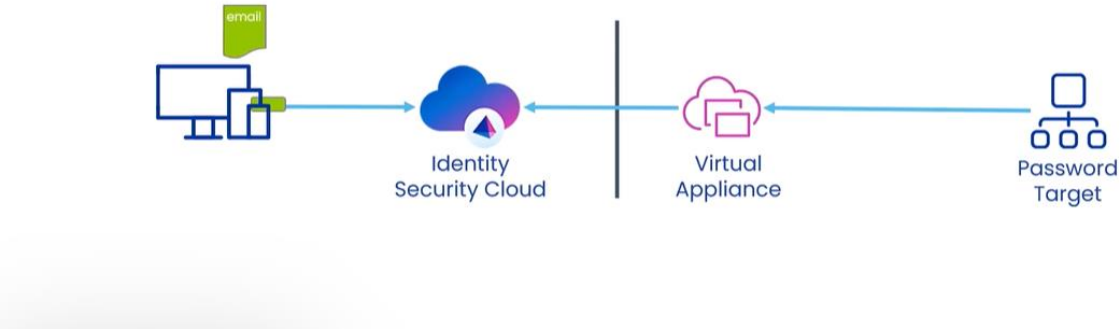
# Password Management

**Password Management Add-on**

Identity Security Cloud Password Management simplifies password administration and updates passwords across Identity Security Cloud account sources and applications. It can be used to enforce consistent and strong password policies within your organization and is an optional Identity Security Cloud add-on.

**Password Change Process**

Password change is the process flow initiated when a user has authenticated to Identity Security Cloud. Once a user begins a request to change a password, what process does Identity Security Cloud follow to verify the request before completing the change?




**Virtual appliance administration**

As an identity security admin, it's important to understand how your tenant communicates with systems in your network. This communication happens through a virtual appliance (VA), a key component that acts as a secure gateway between your tenant and your enterprise systems.

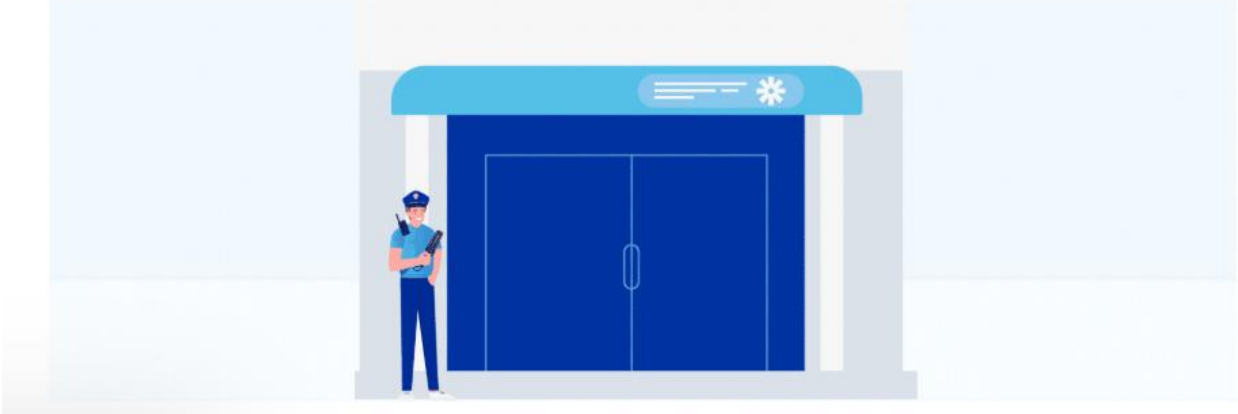
## The condo building

This condo building represents your organization's multi-tenant IT environment, where different tenants (users or groups) live securely, each with access to their own private spaces and shared resources that need protection.



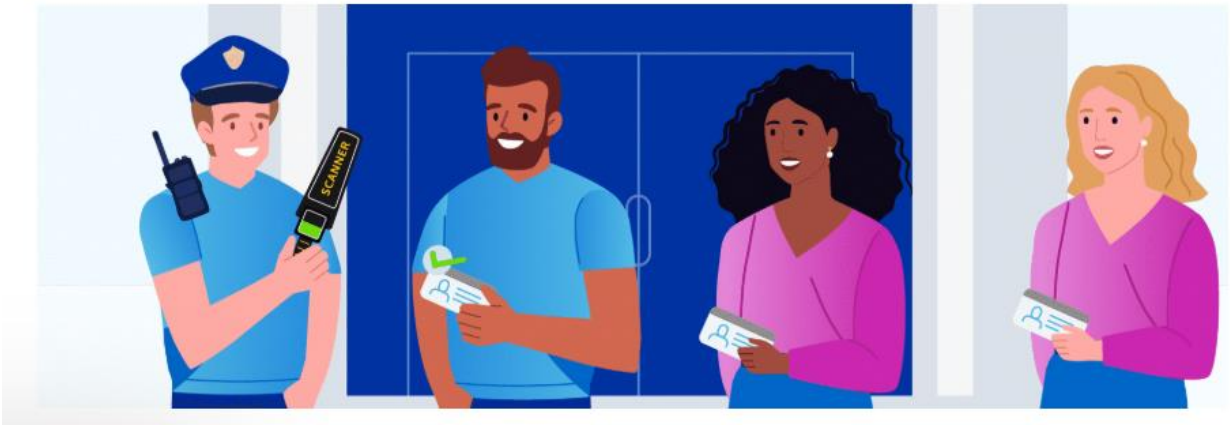
## The digital security guard

The digital security guard symbolizes the SailPoint virtual appliance, managing who can enter the building and ensuring only authorized tenants and visitors gain access to specific units.



## Checking credentials

Just like the guard verifies tenant badges or visitor passes, the VA checks identity and access policies to confirm who is allowed inside and which areas they can access.



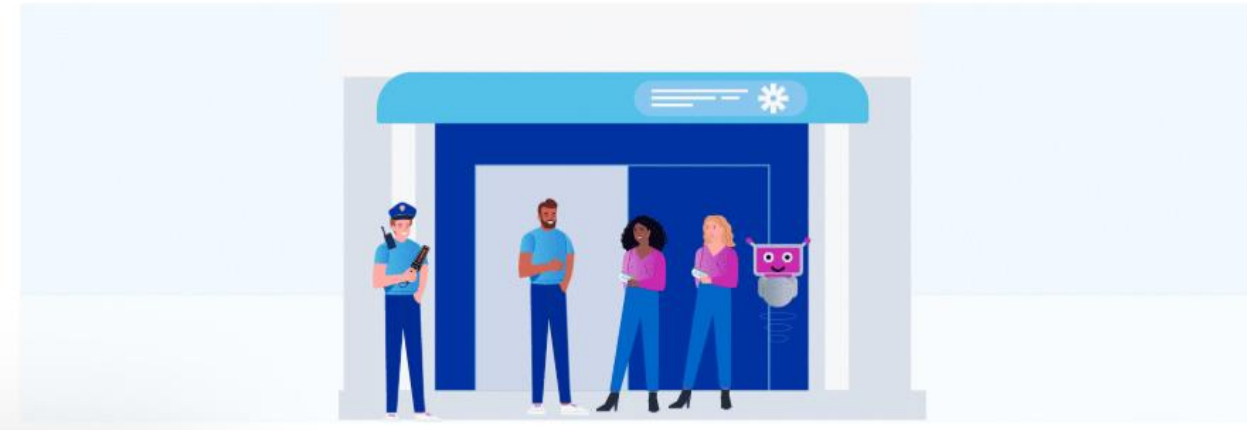
## Using advanced security tools

The guard's tools represent SailPoint's advanced features, policy enforcement, authentication, and access control, that keep the condo building secure and streamline entry for approved tenants.




## Streamlining secure access

Just as identities with valid credentials enter their units without hassle, SailPoint ensures authorized users access the right digital resources efficiently and securely.



### Keeping identity security in check

SailPoint's VA acts as the condo's trusted gatekeeper, maintaining strong identity security and preventing unauthorized access to the community and private spaces.



## Communication flow

Now you'll take a closer look at how communication flows within a virtual appliance cluster. Select each tab below to explore how the VA cluster handles messaging, from the cluster queue and message decryption to the internal communication that keeps everything running securely and efficiently.

**VA cluster**

- The tenant communicates with your VAs by adding messages to the VA Cluster Queue for the VA Cluster which manages the system it's trying to reach
- The VA cluster queue within your tenant is a holding area of messages waiting for the VAs to process

**Cluster queue**

- The VAs defined in the same cluster are constantly polling the same cluster queue for new work

**Cluster queue messages**

- The scheduled activities and end-user actions targeted for specific sources result in messages being added to the appropriate cluster queue

**Message decrypt**

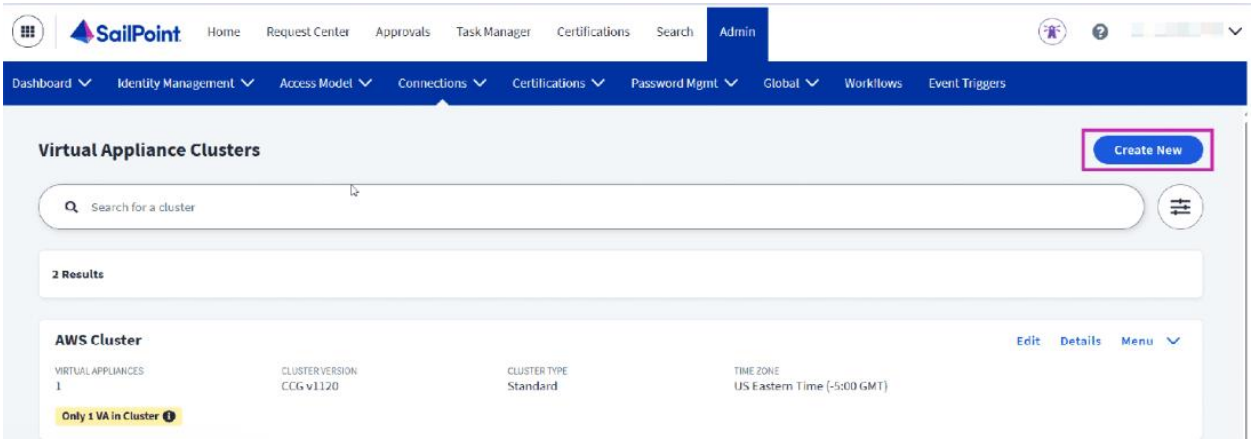
- If a VA finds a message, it pulls down that one piece of work and decrypts the message using the local private key

**Internal communication**

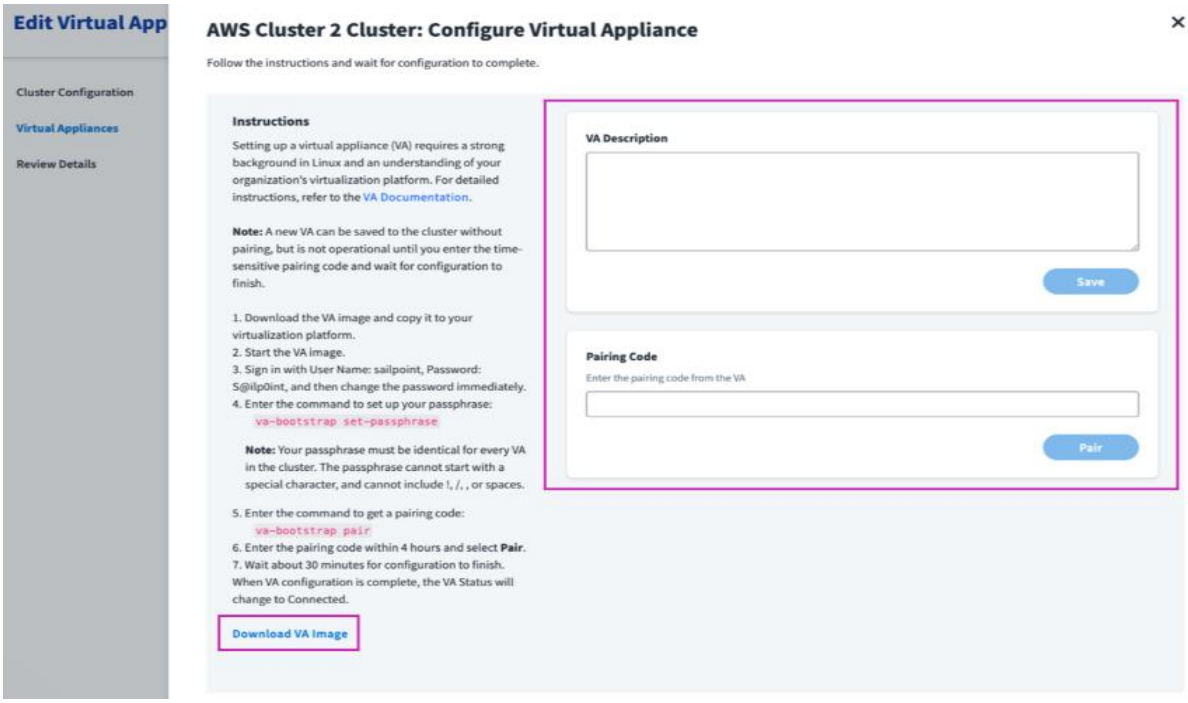
- The VA communicates internally with the cloud connector gateway (CCG) to complete the work
- Only one VA in a cluster will respond to a message at a time
- The completion status and results flow back to the tenant from the VA

# Deployment process

You can set up as many VAs as needed to support your sources, and the user interface provides download links along with clear instructions to guide you through the setup. You'll also see how the latest enhancement, a pairing code for authentication, adds an extra layer of security during deployment.




- The download link on the left pane allows you to pull down the VA image file
- The Process Summary describes the steps required to complete the installation of the VA
- The newest element for this process is that SailPoint now uses a pairing code for authentication


Edit Virtual App
Cluster Configuration
Virtual Appliances
Review Details
AWS Cluster 2 Cluster: Configure Virtual Appliance
Follow the instructions and wait for configuration to complete.
Instructions
Setting up a virtual appliance (VA) requires a strong background in Linux and an understanding of your organization's virtualization platform. For detailed instructions, refer to the VA Documentation.
Note: A new VA can be saved to the cluster without pairing, but is not operational until you enter the time-sensitive pairing code and wait for configuration to finish.
1. Download the VA image and copy it to your virtualization platform.
2. Start the VA image.
3. Sign in with User Name: sailpoint, Password: S@ilp0int, and then change the password immediately.
4. Enter the command to set up your passphrase:
va-bootstrap set-passphrase
Note: Your passphrase must be identical for every VA in the cluster. The passphrase cannot start with a special character, and cannot include !, /, , or spaces.
5. Enter the command to get a pairing code:
va-bootstrap pair
6. Enter the pairing code within 4 hours and select Pair.
7. Wait about 30 minutes for configuration to finish. When VA configuration is complete, the VA Status will change to Connected.
Download VA Image
VA Description
Save
Pairing Code
Enter the pairing code from the VA
Pair

# Connectivity

Connectivity is the backbone of any identity governance strategy, as it's what allows Identity Security Cloud (ISC) to interact with the applications and data that power your organization. ISC's connectivity framework ensures that it can reach out to all your critical systems, gather the necessary identity data, and enforce access policies, whether those systems are in the cloud or on your own servers.

**Sources**

The foundation of any integration within ISC is the source. A source represents an external resource, like an application, database, or directory management system, that maintains user accounts or personnel data. With source functionality in place, organizations can use ISC to connect their various sources and user data sets and manage access across all those different sources in a secure, scalable way.

**Represents an application within ISC**

Administrators create sources in ISC, where each source represents one of the organization’s applications.

**Stores user attributes (names, emails) and access details**

Attributes may contain names, usernames, emails, departments and so on, as well as data restrictions or accesses.

**Enables aggregation (data import) and provisioning (change export)**

- Aggregation reads data from each system
- ISC provisions attributes and access changes back to the external applications

Some examples of sources include your organization's ERP and financial systems, customer management systems, HRIS systems, Active Directory, Microsoft Entra, GitHub, custom applications, JDBC databases, and even spreadsheets that contain user data that needs to be governed.

# Connectors

Connectors are pieces of code logic that act as a bridge between the SailPoint ISC SaaS platform and the source systems that ISC communicates with and aggregates data from. For instance, ISC synchronizes data between systems, such as Oracle HR or GitHub, to ensure account entitlements are correct throughout your organization.

## The importance of connectors

**Aggregation**

An inbound process where user account and access data are collected from sources

**Provisioning**

An outbound process where account and access changes to sources are suggested

**Integration**

A deeper functionality with service desk and collaboration applications

**IQService**

IQService is a lightweight integration service that enables ISC to participate in a Windows environment. It creates a communication channel for data that is only available through Windows APIs.

Here are some key considerations when using IQService:

**Installation**

IQService must be installed on all Windows servers with connectivity to target systems.

**Security**

It's not mandatory to have client authentication for this communication channel. However, to further secure the communication channel, certificate-based client authentication can be configured.

**Supported sources**

- Microsoft Active Directory
- Entra ID
- SharePoint Online and Server
- HCL Domino
- Windows Local Source

## Common connectors

There are two primary categories of connectors for sources:

**VA-based connectors**

VA-based connectors provide an interface to read user account data and provision changes from ISC to target systems and applications. VA-based connectors are on-premises deployments which require you to configure a virtual appliance cluster to manage the connector. They were previously referred to as "direct sources."

**SaaS connectors**

SaaS connectors provide an interface to read user account data and provision changes from Identity Security Cloud to target systems and applications without needing an on-premises virtual appliance cluster.

# User and application onboarding

As an identity security admin, you'll be involved in onboarding both users and applications, key processes that support identity management and access within ISC. Understanding how these onboarding flows work will help you ensure smooth and secure operations.

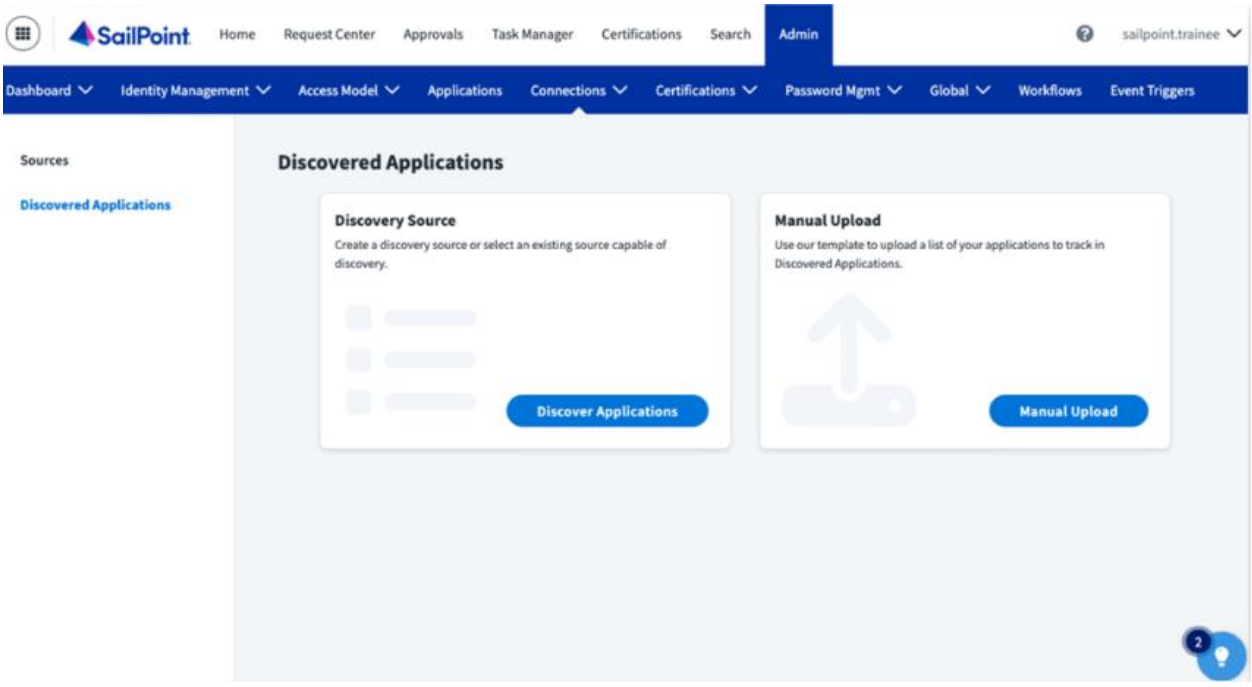
**Source configuration**

The first step to onboarding users to ISC is to create and configure sources. From there, you can use the aggregation process to load identity data into tenants. In general, there are various types of data sources, but the important thing to remember is that they represent the access you want to manage, ensuring reliable and secure access to ISC.

**Application onboarding**

Application onboarding involves two key steps: first, discovering the right sources; and second, getting configuration recommendations so you can manage enterprise applications using ISC.

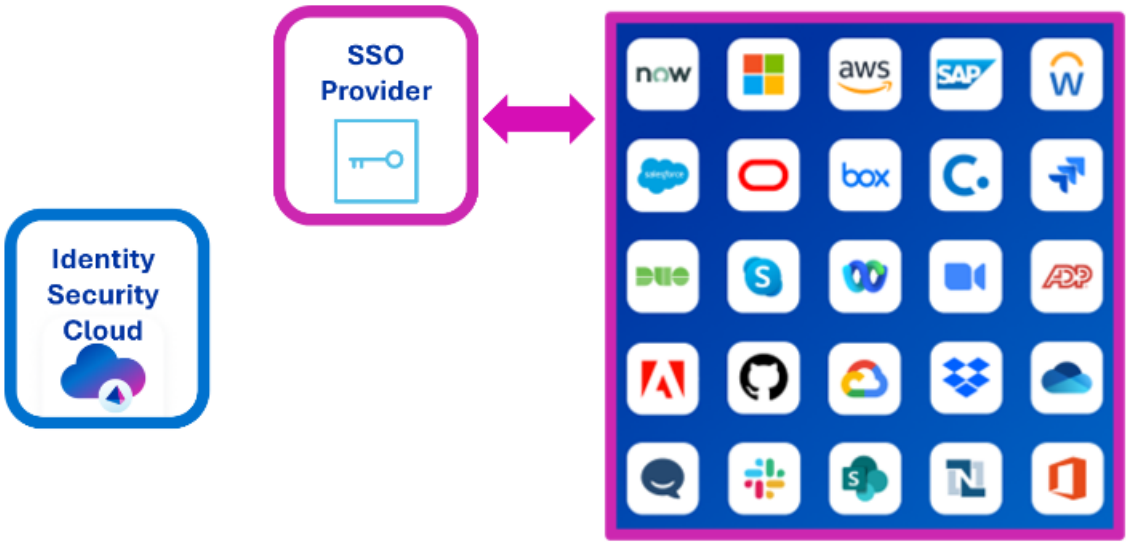
When you're ready to bring your applications into ISC and create sources, the application onboarding feature is your guide. It simplifies the process of onboarding applications, creating sources, and setting up configurations.




**Discovering applications automatically**

## 01 Step 1

To discover applications, we first look at what's already in place. In this example, you'll see that many business applications are already configured to use a single sign-on (SSO) provider. Understanding this existing setup is the first step.




## 02 Step 2

You then connect ISC to the SSO Provider.



## 03 Step 3

Next, you'll use AI application onboarding in ISC to help identify the connectivity details needed for the applications you've set up in the SSO provider.




## 04 Step 4

This information is then loaded into ISC.



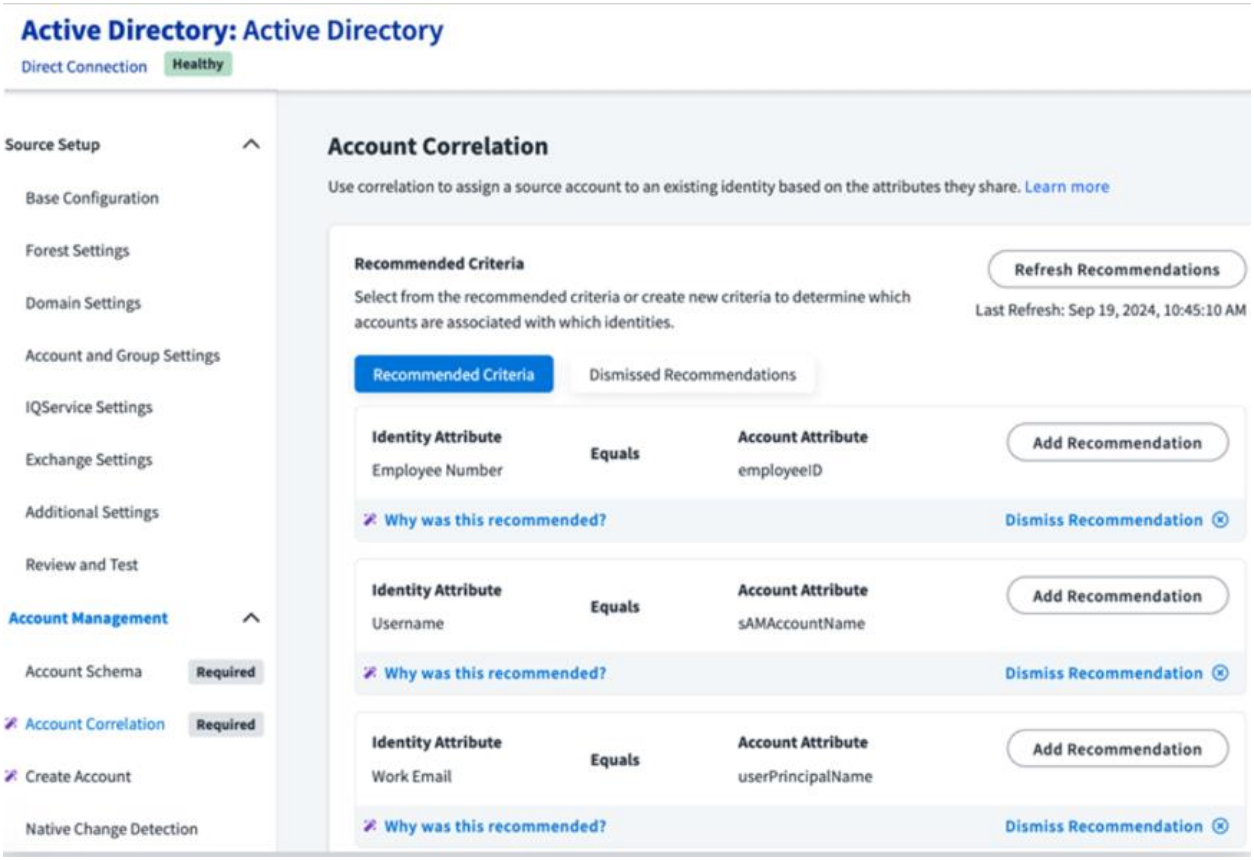
## 05 Step 5

ISC can now connect directly with those business applications.



# Application onboarding configuration

Configuration recommendations aim to reduce setup complexity, specifically around account schema and correlation logic. After applications are discovered, either automatically with AI or manually, the goal is to make source creation and the overall onboarding process more efficient for identity security admins.



# Onboarding Identities

What is an identity?

An identity is the governed digital representation of a person or non-person access-holding entity within ISC. It serves as the central aggregation point for governance data, including authoritative identity attributes, correlated accounts across target systems, associated entitlements, roles, and access relationships. The identity enables controlled access to applications, systems, networks, and data in accordance with organizational policies.

## Onboarding identities (Authoritative) implementation steps

Onboarding identities lays the groundwork for secure and effective identity governance through seven implementation steps, each designed to align identities with your organization's access policies and systems.



## Data cleaning

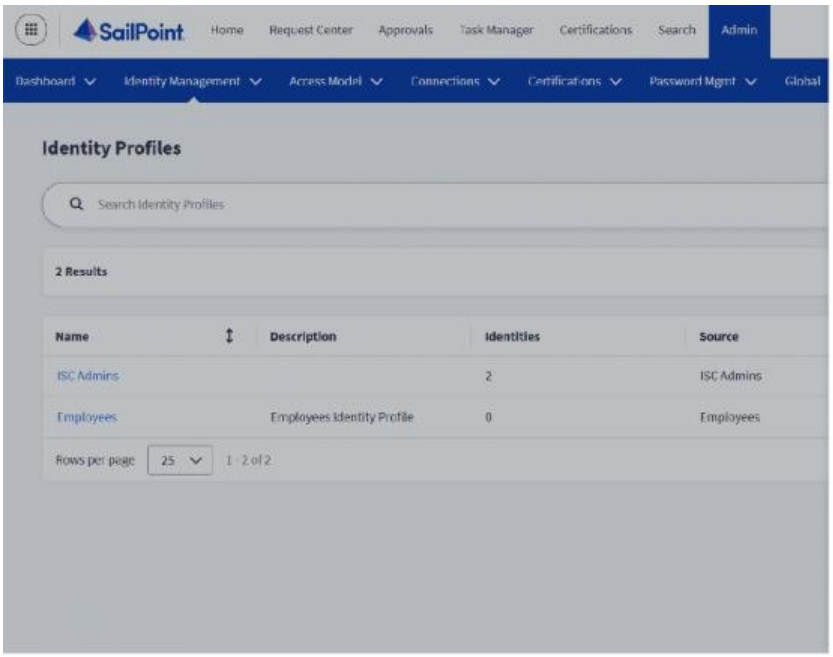
The first step in onboarding identity data is to ensure your data is **clean**, meaning it meets SailPoint's required standards. Clean data is the foundation of identity governance and AI recommendations.

## Source configuration

ISC collects information about your users' system accounts and their associated access so they can be governed. Once you configure a source, you can manage how your data is aggregated, or loaded, in your environment.

## Identity profile configuration

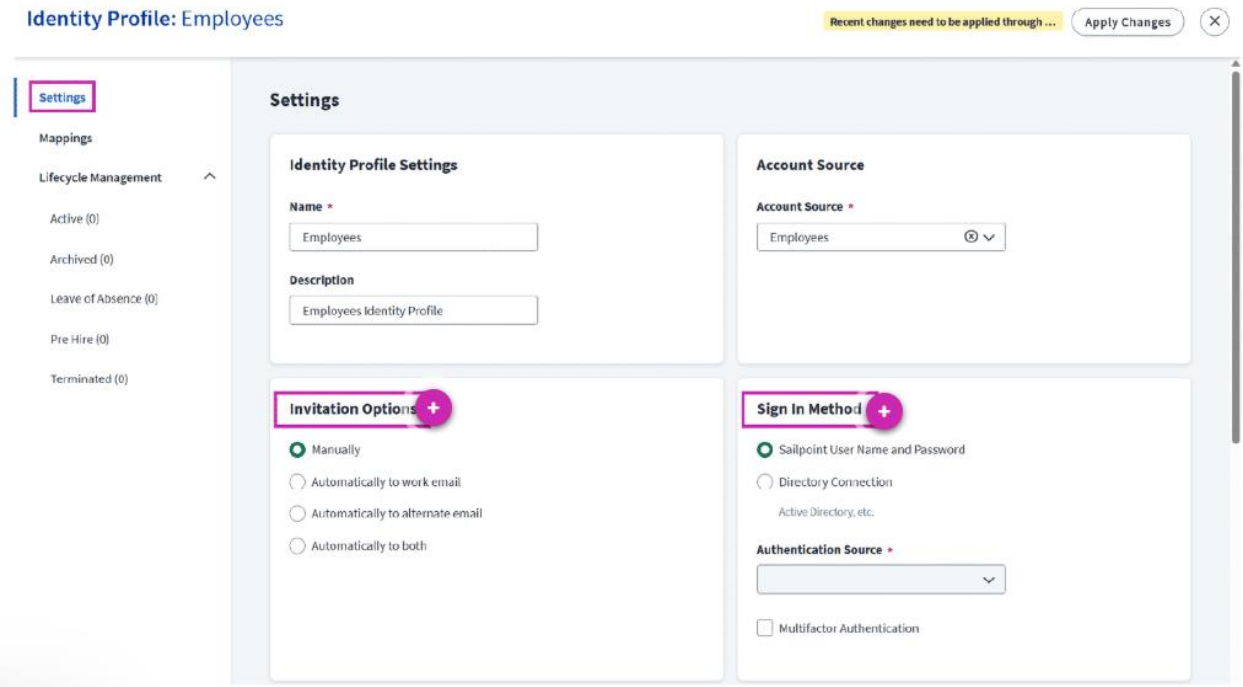
The next step of the onboarding identities process focuses on configuring your authoritative source by creating an identity profile and configuring manager correlation. Setting up an identity profile for a source tells ISC that the source is authoritative for creating identities and that new accounts read from that source should define new identities for your organization

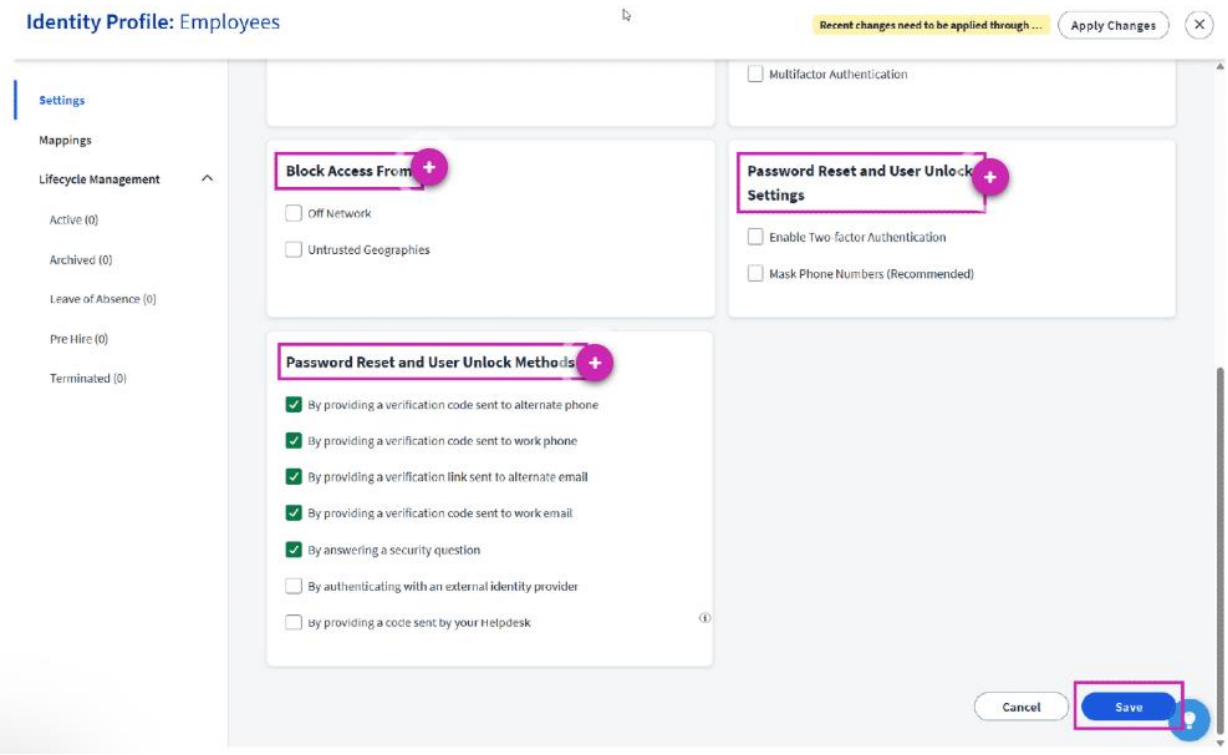





## Settings

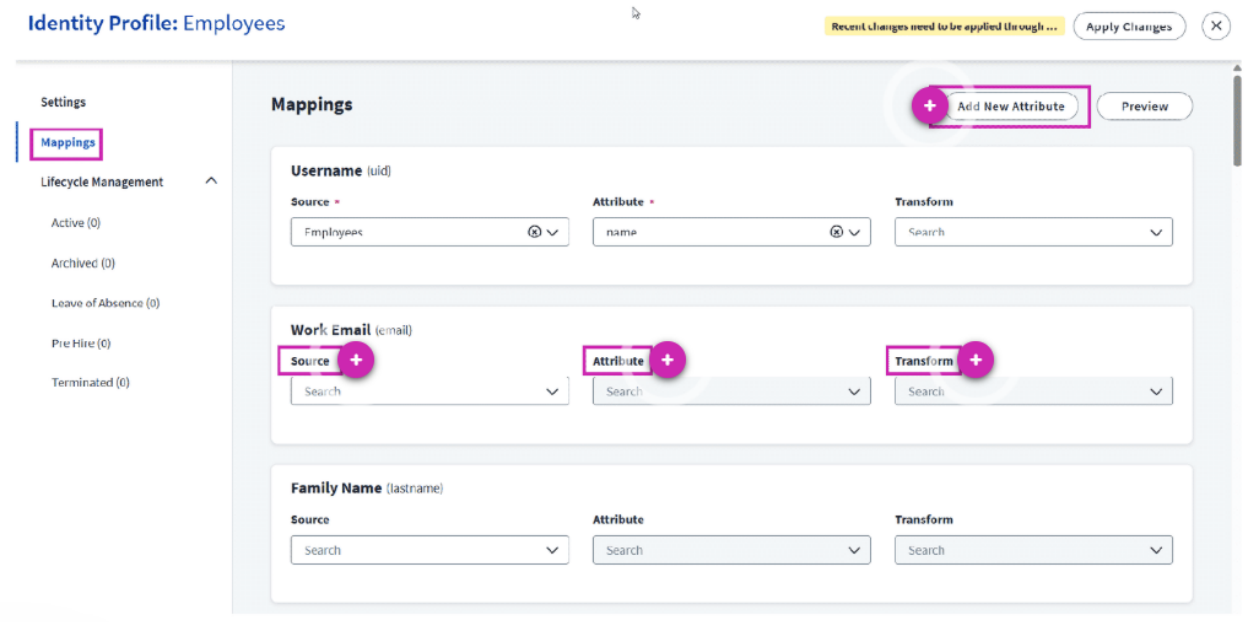
Configure the identity profile's sign-in and security settings.






## Mappings

Identity attributes contain information about each identity used to manage its access. The identity profile defines the attributes to be used for the set of identities created from that profile and where each attribute of the identity will get its value.

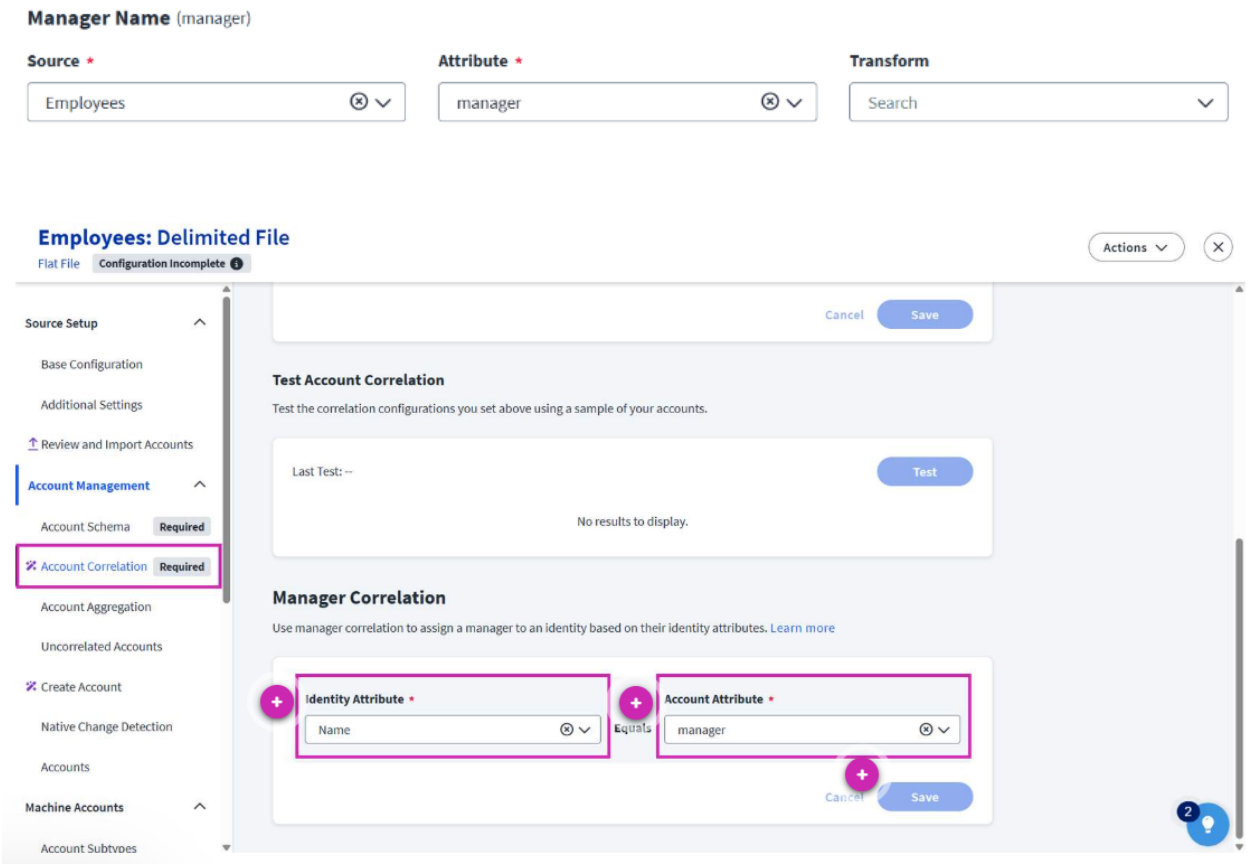


**Lifecycle Management**

Lifecycle States define employment relationships with an organization, which you can use to drive access changes for your users. For example, when a new employee joins your company, ISC can grant them the required access for active employees. When someone leaves the organization, their access can be automatically revoked or their source accounts are disabled.

**Manager correlation**

After establishing your account schema and identity profile, you must configure Manager Correlation in your authoritative source configuration. Manager correlation links an employee to their manager. This relationship is used for certifications (to review and certify users' access, which is covered in the *Governance* module) and access request approvals.




# Account aggregation

In ISC, account aggregation is the process of reading data into the cloud from a source using the connectivity information and schema specified in the source definition (which applies to all source types). For each account on the source, the attributes named in the schema are imported and recorded in an account record in the cloud.

**Verify data cleanliness (by addressing identity exceptions)**

The next step in the onboarding identities process focuses on verifying the cleanliness of your identity data by addressing exceptions and errors. Let's take a closer look at how this is done.

**Troubleshoot identity exceptions**

There are two identity exceptions encountered when onboarding:

- **Incomplete identities:** These occur when ISC detects bad or missing data during identity creation; some examples are:
    - A non-unique username (uid)
    - Missing last name, work email, or username
- **Identities with errors:** These are usually the result of a bad transform operation

When exceptions occur, ISC will provide access to them from the Identities Lists (**Admin** > **Identity Management** > **Identities**)

# Onboarding identities (non-authoritative sources)

**Configure non-authoritative sources**

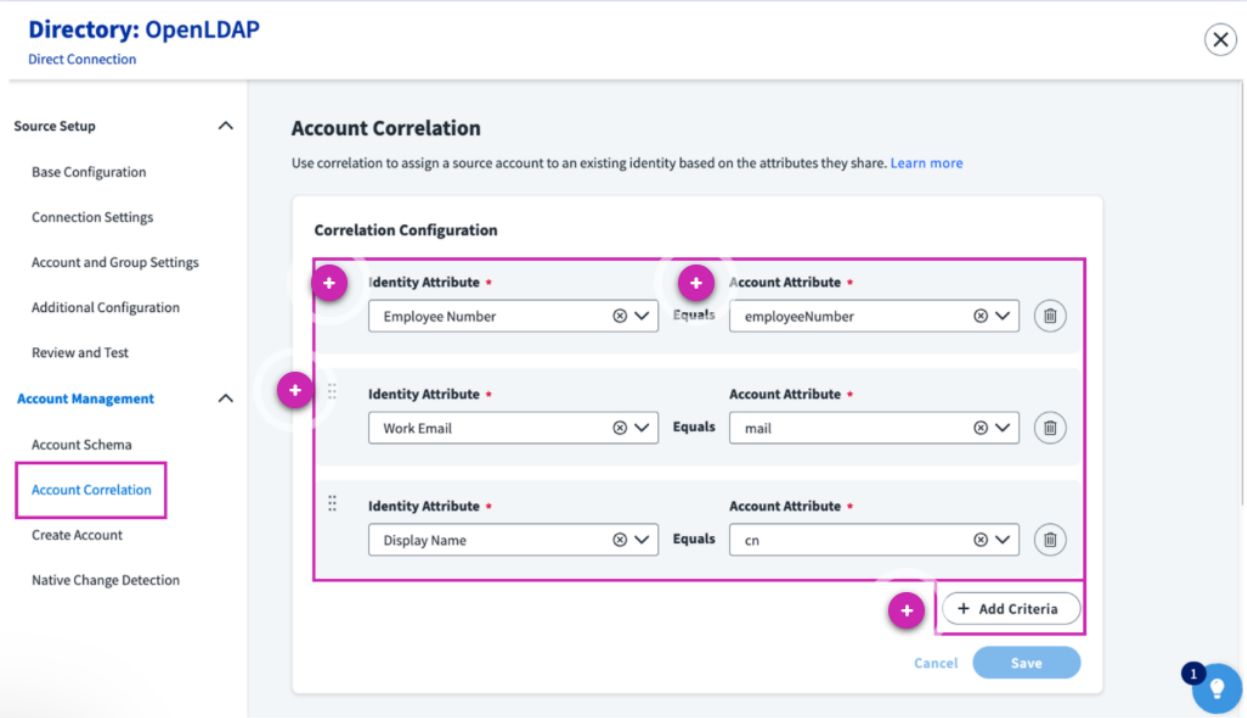
The fifth step in onboarding identity data is to configure non-authoritative sources, which are external applications, databases, or directory management systems that manage their own sets of users but are not considered the primary system of record. Proper configuration ensures accurate and reliable identity data will be loaded into Identity Security Cloud (ISC).

**Account correlation**

Configuring and aggregating non-authoritative sources and entitlements begins with account correlation. Correlation is the process of matching and assigning source accounts to identities. Correlation configuration compares the values of specific account attributes with the values of related identity attributes. When those attribute values match, the account is assigned to the identity.

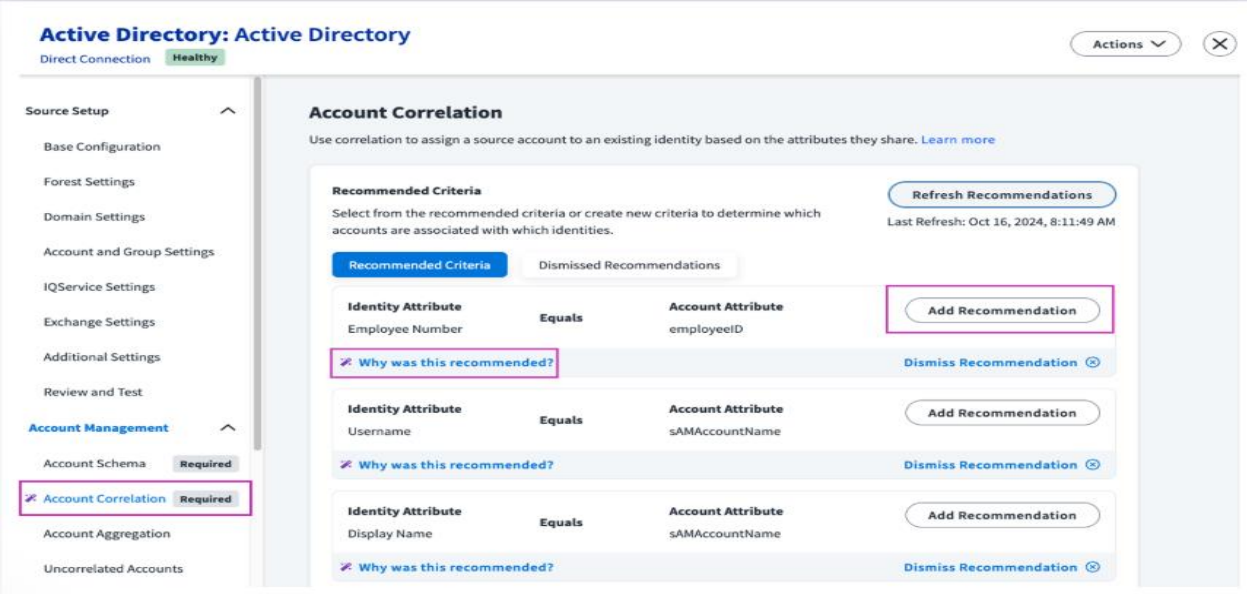
**Correlation configuration**

Correlation configuration is the process of setting up rules to automatically link related data or events. To correlate accounts to identities, you will map, or pair, account attributes to the identity attributes in your tenant. These mappings are configured using correlation criteria, which compare the values of account and identity attributes. Whenever one correlation criteria is met, the account is correlated to the identity.



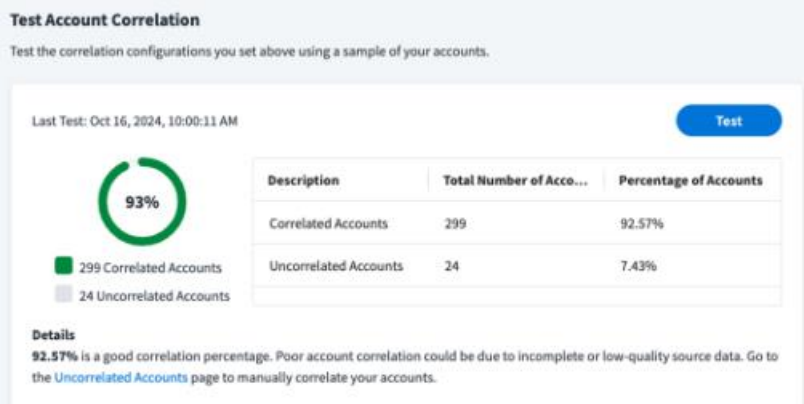

# AI-Driven correlation recommendations

If your organization uses recommendations, you may see suggested criteria to select based on correlation percentages. These suggestions depend on the accuracy of your identity and account data. Use them to set up, refine, and test account correlations, helping accounts match the correct identities more easily.

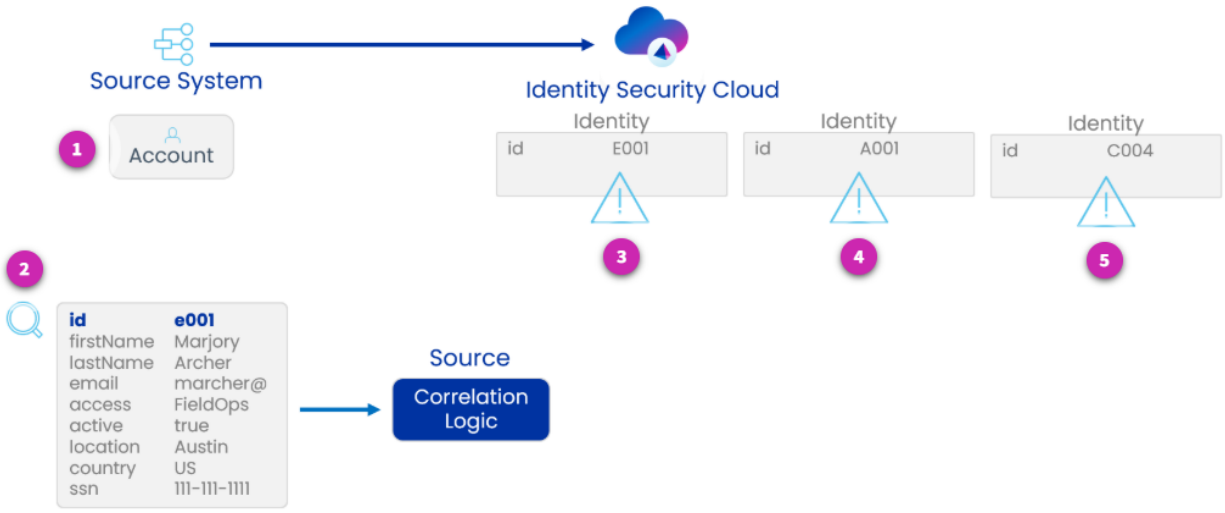


## Correlation testing

When using **Test Account Correlation**, ISC checks the correlation configuration against a sample of your accounts to estimate the number and percentage of your accounts that will be correlated.




**Correlation example**




**Solving uncorrelated accounts**

If ISC is unable to match account attributes to identity attributes, the account is considered uncorrelated and is not assigned to an identity. Accounts might be uncorrelated for a number of reasons. Some accounts simply need to be removed from a source, while others require updates to a source or to information in your system. For example, one record might use a nickname while another uses a formal name. Or an account might belong to a former employee.

**Resolve uncorrelated accounts on sources**

To resolve mismatched attribute values across multiple sources, you can use the global Uncorrelated Accounts report:

1. Go to **Admin** > **Global** > **Reports**
2. Select **Actions** > **Generate .pdf** or **Generate .csv** in the **Uncorrelated Accounts** row to generate a list of all uncorrelated accounts
3. Download the report and review the list of uncorrelated accounts and their sources

uncorrelated-accounts 20210517190815

| Identity | Account | Source |
|---|---|---|
| Bob.Brown | CN=Bob Brown,OU=Austin,OU=Americas,OU=Demo,DC=seri,DC=sailpointdemo,DC=com | Corporate Active Directory |
| Bill.Moore | CN=Bill Moore,OU=San Jose,OU=Americas,OU=Demo,DC=seri,DC=sailpointdemo,DC=com | Corporate Active Directory |
| Lois.Lane | CN=Lois Lane,OU=Singapore,OU=Asia-Pacific,OU=Demo,DC=seri,DC=sailpointdemo,DC=com | Corporate Active Directory |
| SRV-IISService | CN=SRV-IISService,OU=San Jose,OU=Americas,OU=Demo,DC=seri,DC=sailpointdemo,DC=com | Corporate Active Directory |
| Enterprise 219 | d80aab0f-f505-4fb5-aa4e-6408fe65258f | Azure AD |
| admin | fe83ee97-2e8a-41cb-9dd6-53cd13f9dc57 | Azure AD |
| SRV-SQLServerDaemon | CN=SRV-SQLServerDaemon,OU=San Jose,OU=Americas,OU=Demo,DC=seri,DC=sailpointdemo,DC=com | Corporate Active Directory |
| csimmons-adm | CN=csimmons-adm,OU=San Jose,OU=Americas,OU=Demo,DC=seri,DC=sailpointdemo,DC=com | Corporate Active Directory |
| Aaron.Nichols | CN=Aaron Nichols,OU=Singapore,OU=Asia-Pacific,OU=Demo,DC=seri,DC=sailpointdemo,DC=com | Corporate Active Directory |
| Pamela.Kelly | CN=Pamela Kelly,OU=San Jose,OU=Americas,OU=Demo,DC=seri,DC=sailpointdemo,DC=com | Corporate Active Directory |
| harper_test | CN=harper_test,OU=Singapore,OU=Asia-Pacific,OU=Demo,DC=seri,DC=sailpointdemo,DC=com | Corporate Active Directory |

4.On the source listed in the report, edit at least one attribute on each uncorrelated account, including the value of an attribute used for correlation

- For example, if you're editing an account in Active Directory, you can update the sAMAccountName to match the uid of an identity in your system, if those attributes are paired in the correlation configuration

5.Save your changes and aggregate the source's accounts again

## Perform a manual correlation

You can use manual correlation to resolve uncorrelated accounts by creating a permanent link from the account to the identity. Even if the source data for the account changes, the account will still be linked to the identity. To correlate the account to a different identity after it was manually correlated, you must manually remove the account from the identity.

To resolve uncorrelated accounts manually, complete these steps:

1. Go to **Admin** > **Connections** > **Sources**
2. Select or edit the source with uncorrelated accounts
3. In the Account Management section, select **Uncorrelated Accounts**
4. Select **Download Correlation File** to download a .csv file of uncorrelated accounts
5. In the .csv file, add the Account IDs of the associated identities to the userName column beside the uncorrelated account

6.When you're finished making changes, save the file

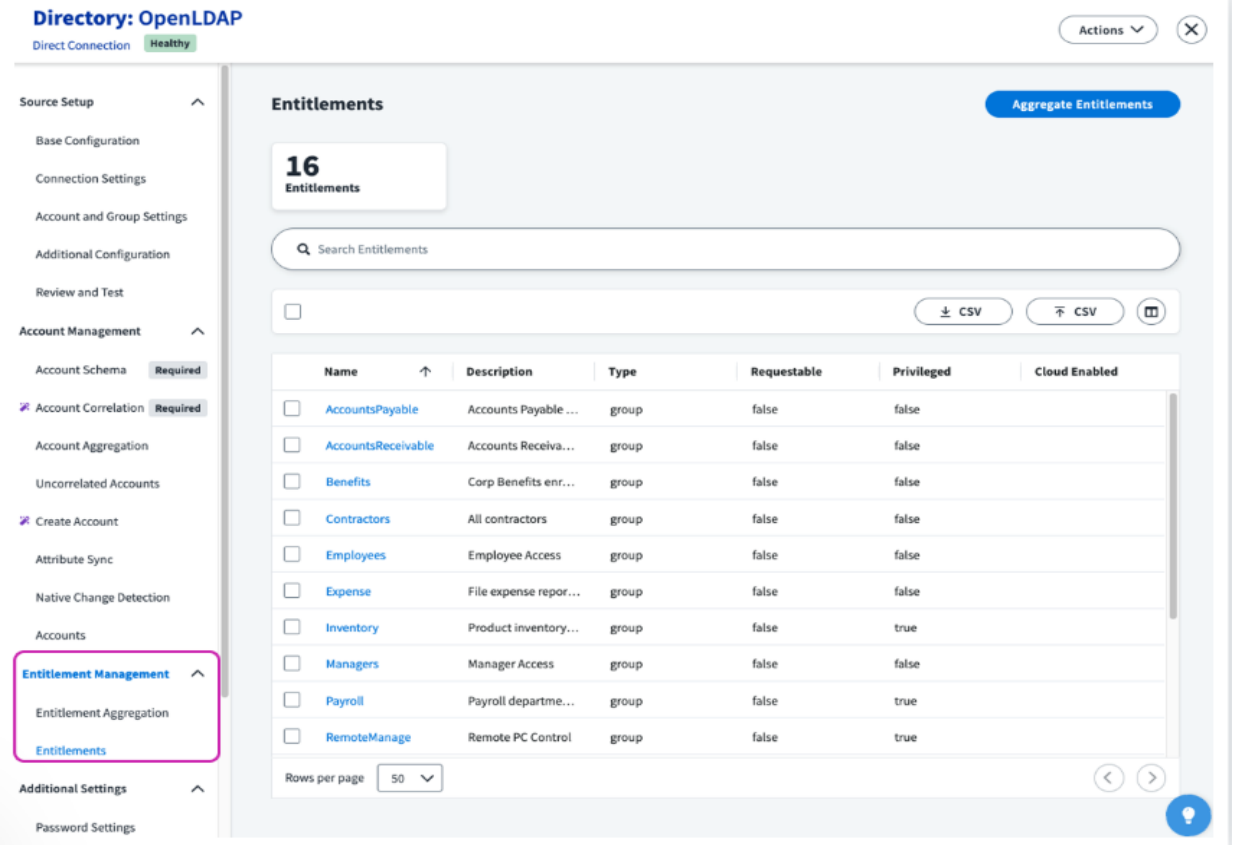
7.On the **Uncorrelated Accounts** page, upload the .csv file

## Entitlement aggregation

Understanding and aggregating entitlements is essential for managing user access across connected systems

**Aggregate non-authoritative accounts and entitlements**

The sixth step in onboarding identity data is to aggregate accounts and entitlements from non-authoritative sources. This process brings in additional access data that helps build a complete picture of user permissions across your environment.

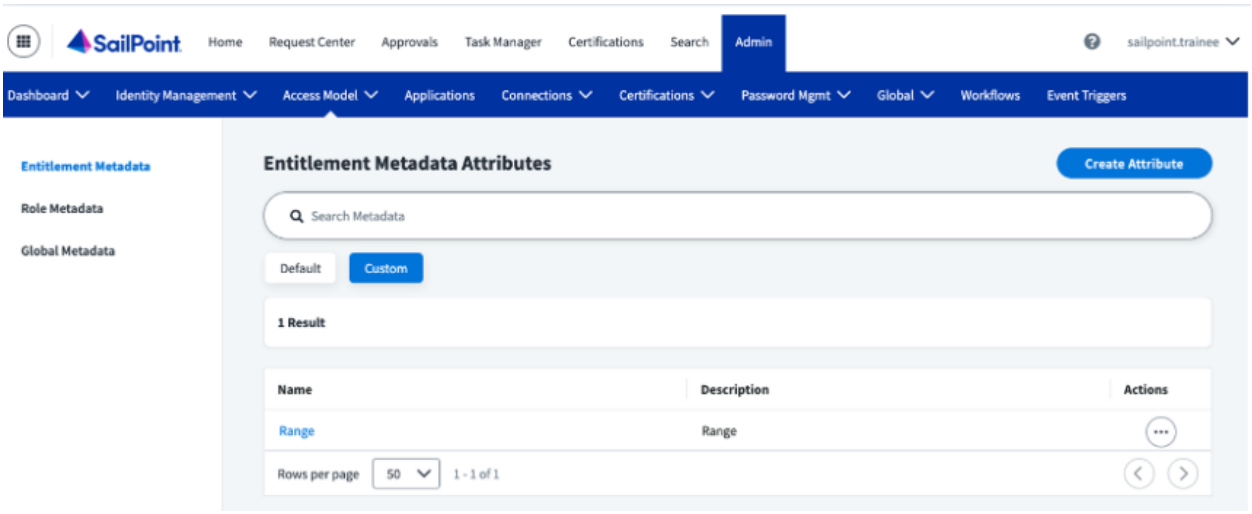



Having access to the full entitlement catalog allows you to:

- Model and combine entitlements in ways that support your access policies
- Prepare for provisioning, access requests, and governance
- Identify and mark sensitive entitlements as privileged

**Entitlement metadata**

Entitlement metadata can help you make entitlements clearer and easier to manage, adding extra details or context is key

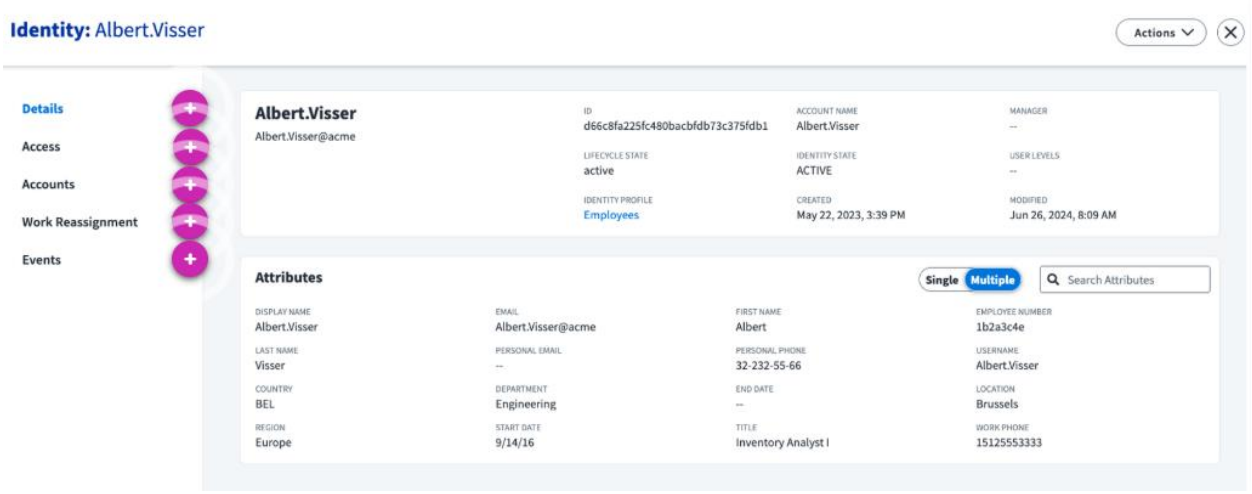



## Reviewing identity data

Ensuring the accuracy and completeness of synchronized identity and account data is critical for maintaining a reliable identity governance program.

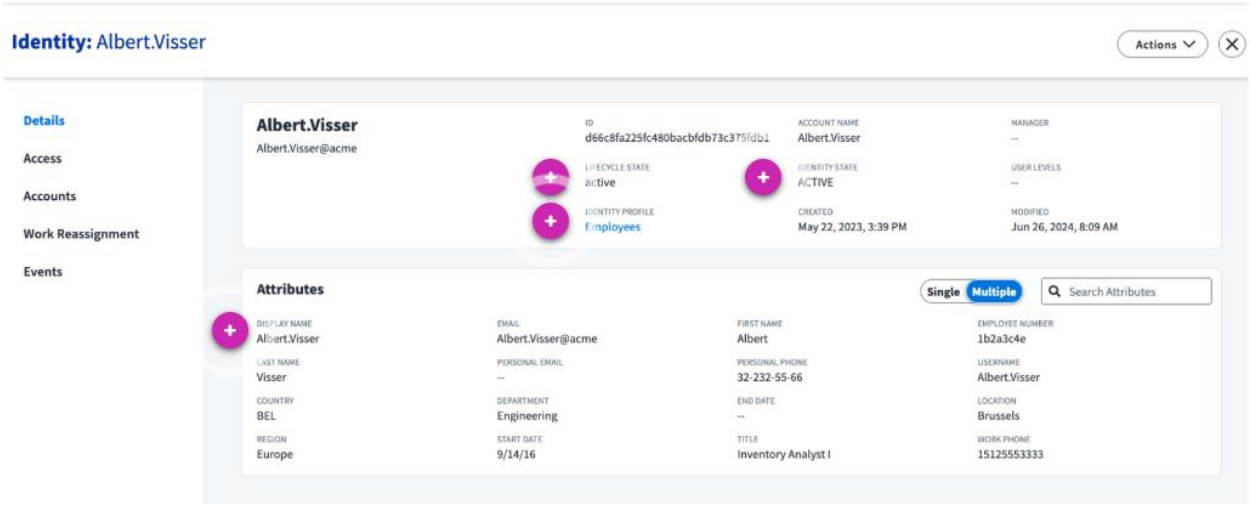
**Verify data cleanliness (by reviewing identity data)**

The final step in onboarding identity data is to verify the cleanliness of synchronized identity and account information. Reviewing this data ensures it is accurate, complete, and aligned with source systems.



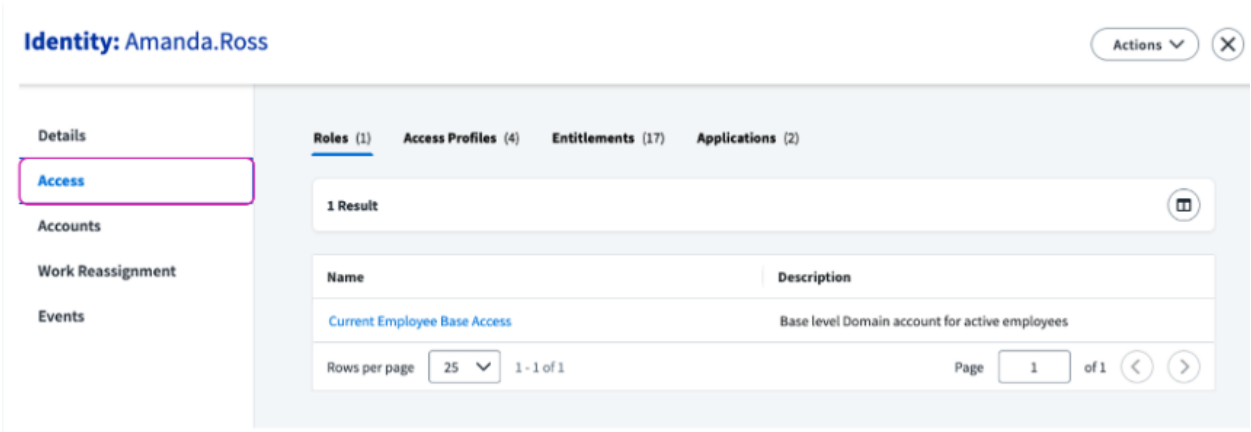
# Details

The **Details** tab displays all identity attributes for a user, such as **Albert**. These attributes are divided into two sections:




## Access

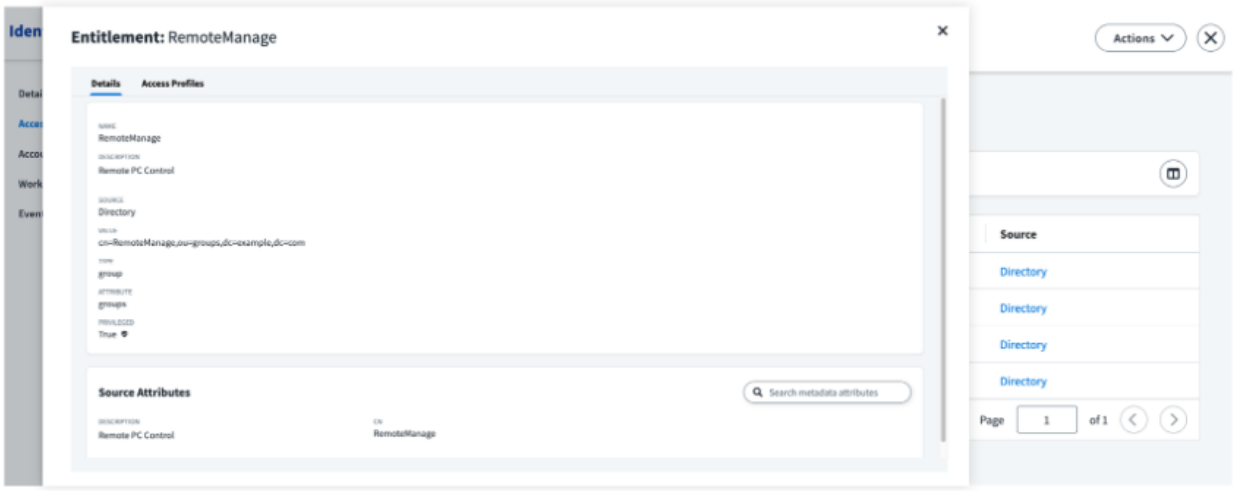
Located just below the Details tab, the Access tab includes important assigned access for the user, Amanda.




Looking deeper into one of the assigned access sections, you land on Entitlements. Access entitlements are like keys that unlock precise capabilities within systems.

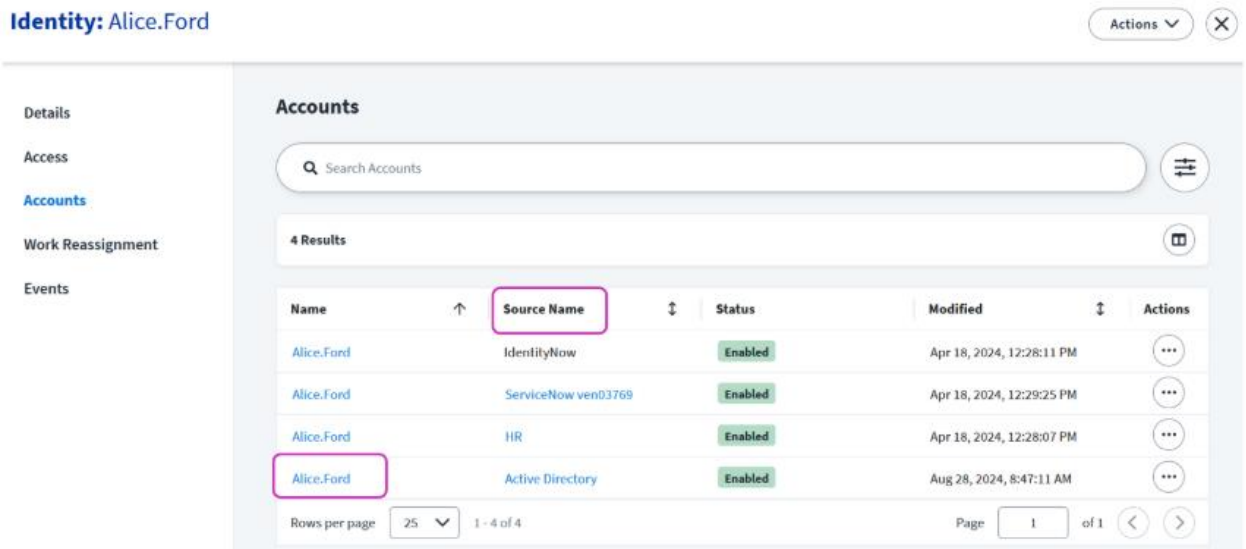
By clicking the Entitlement tab, details regarding available entitlements will be displayed, at a minimum the name, and possibly more.

Once you understand the access held, you can view the account details.




## Account

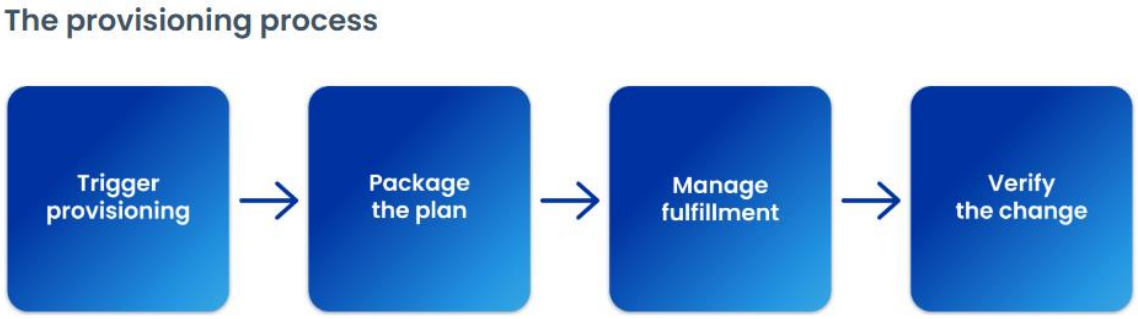
Once user accounts have been aggregated from various sources, each account appears under that person's identity in the **Accounts** section. On this page, you'll see the name of each account and a menu of actions you can take such as enabling, disabling, unlocking, or removing the account.



# The provisioning process

**What is provisioning?**

Provisioning is the process of granting, changing, or removing a user's access to systems, applications, and databases based on their identity profile and the configuration rules for the source. Requests for provisioning can come from various sources and may affect multiple target systems (like your Active Directory or Workday) and types of data.




**Step 1: Trigger provisioning**

Provisioning always begins with a trigger and there are two types of triggers: an automated event or a request-driven event in ISC that causes a data change.

**Step 2: Package the plan**

Triggers create a standard data format that executes the requested provisioning action. All actions are packaged the same way, no matter how they start or what system they target. Each provisioning plan is specific to one identity.

**Step 3: Manage fulfillment**

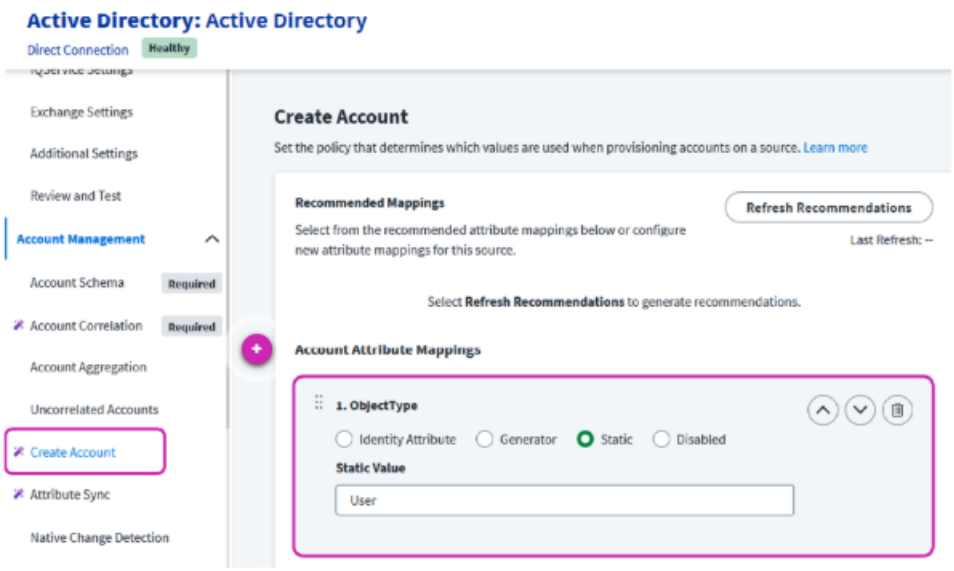
In the manage fulfillment phase, the **actual change** happens (like creating, updating, or removing an account or access). Fulfillments can be managed through different fulfillment channels. These provide the path for sending changes to a downstream system and applying them to the right account.

ISC will automatically check to verify that the selected operation is the correct one. For example, if *modify account* is chosen, but there is no account, it will change to *create account*. Additionally, in this phase, ISC will automatically filter out any access that the identity already has and only provision new access.

**Step 4: Verifying the change**

When a provisioning action is conducted, the connector or integration may send a status update back to ISC. Verification happens after aggregation. Once the account data has been updated, the system can confirm that the provisioning change was successful.

### Configuring source account provisioning



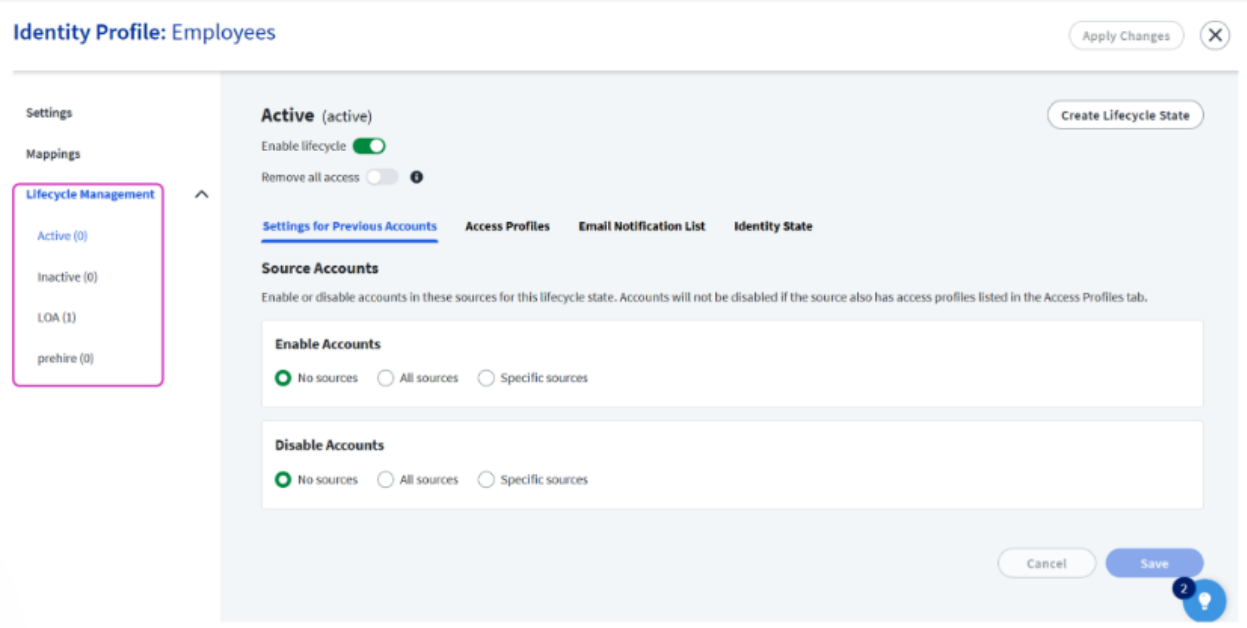

### Passwords in provisioning

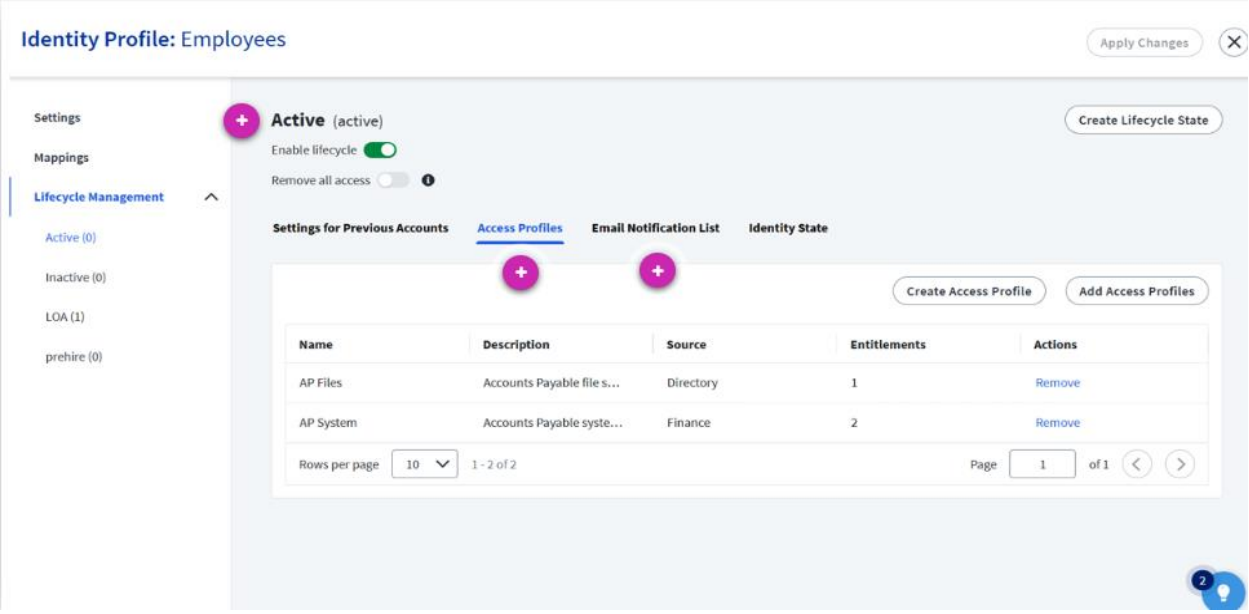
When ISC creates new accounts in provisioning operations, those accounts often need an initial password value. However, as a security design decision, ISC **never** sends out passwords through email.

# What are lifecycle states?

A lifecycle state represents the status that defines the current relationship of an identity to the organization.

Creating lifecycle states

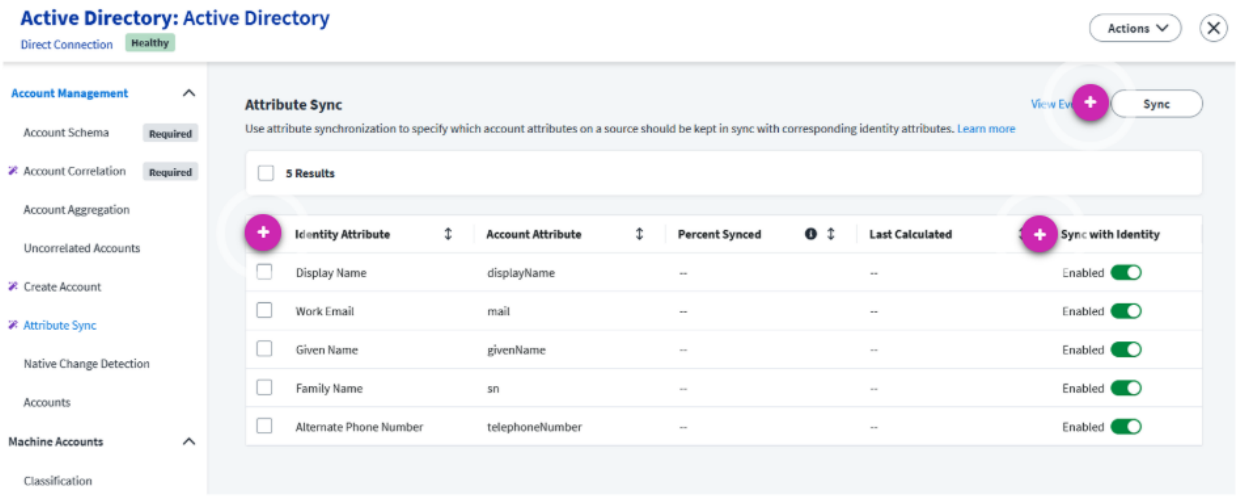


### Synchronizing attributes

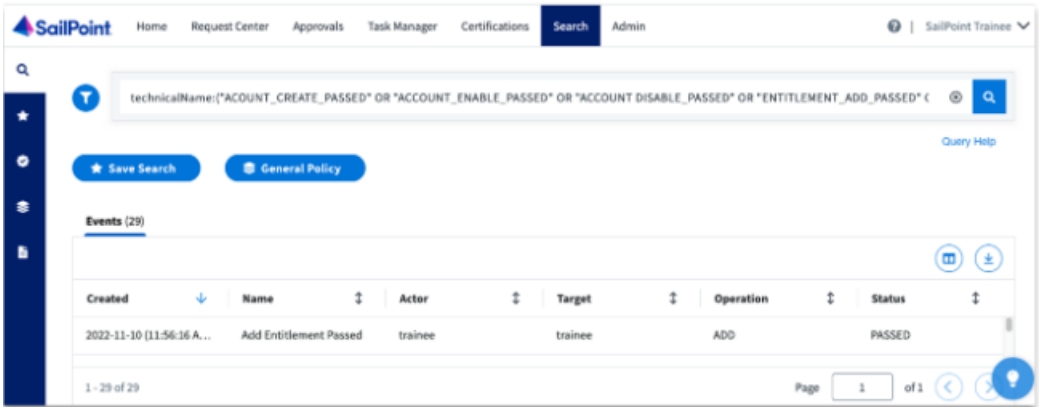
Attribute synchronization (or attribute sync) is the process ISC uses to make sure identity data stays consistent across multiple systems. When something changes in the source system, like a phone number, job title, or department, attribute sync makes sure those updates flow to every connected target system.

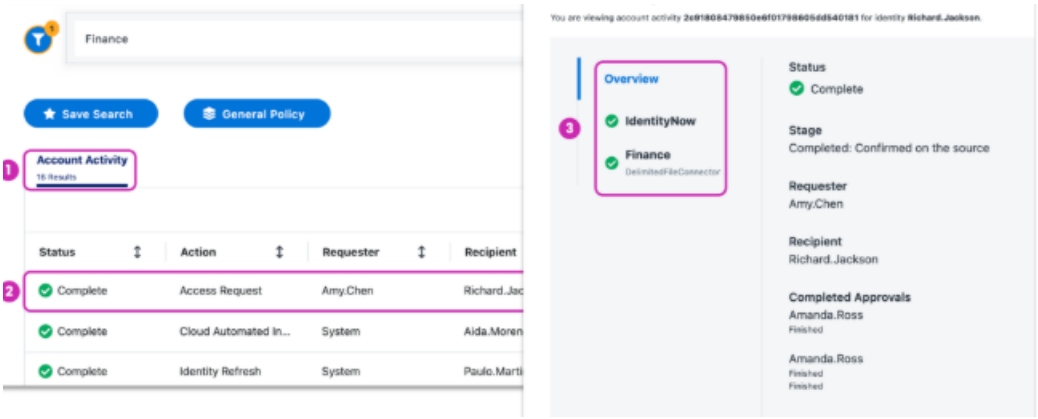
Unlike access changes (granting or removing permissions), attribute sync focuses solely on keeping data accurate and up to date.

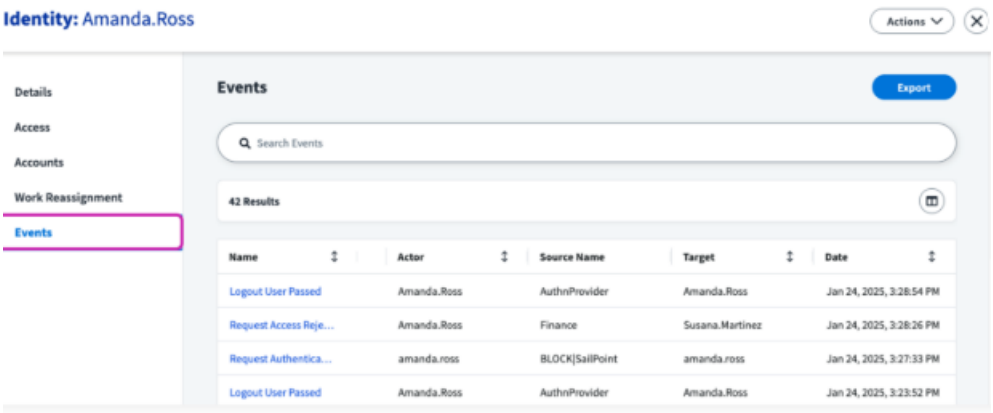


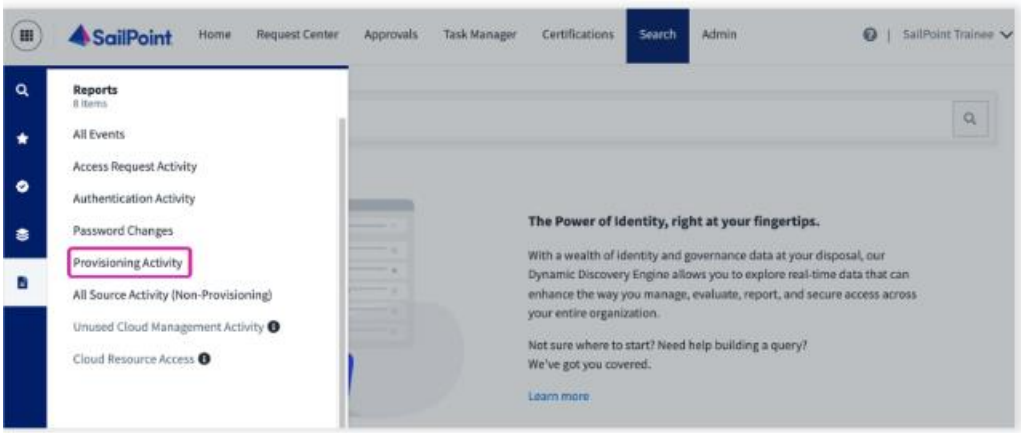

### Monitoring provisioning

As an admin, you have different ways to monitor provisioning actions in the system. ISC provides views showing the status of requests, account changes, and completed operations.




Finance
Save Search
General Policy
Account Activity
Status
Action
Requester
Recipient
Complete
Access Request
Amy.Chen
Complete
Cloud Automated In...
System
Complete
Identity Refresh
System
Overview
IdentityNow
Finance
Status
Complete
Stage
Completed: Confirmed on the source
Requester
Amy.Chen
Recipient
Richard.Jackson
Completed Approvals
Amanda.Ross
Amanda.Ross


Identity: Amanda.Ross
Actions
Details
Access
Accounts
Work Reassignment
Events
Events
Export
Search Events
42 Results
Name
Actor
Source Name
Target
Date
Logout User Passed
Amanda.Ross
AuthnProvider
Amanda.Ross
Jan 24, 2025, 3:28:54 PM
Request Access Reje...
Amanda.Ross
Finance
Susana.Martinez
Jan 24, 2025, 3:28:26 PM
Request Authentica...
amanda.ross
BLOCK|SailPoint
amanda.ross
Jan 24, 2025, 3:27:33 PM
Logout User Passed
Amanda.Ross
AuthnProvider
Amanda.Ross
Jan 24, 2025, 3:23:52 PM

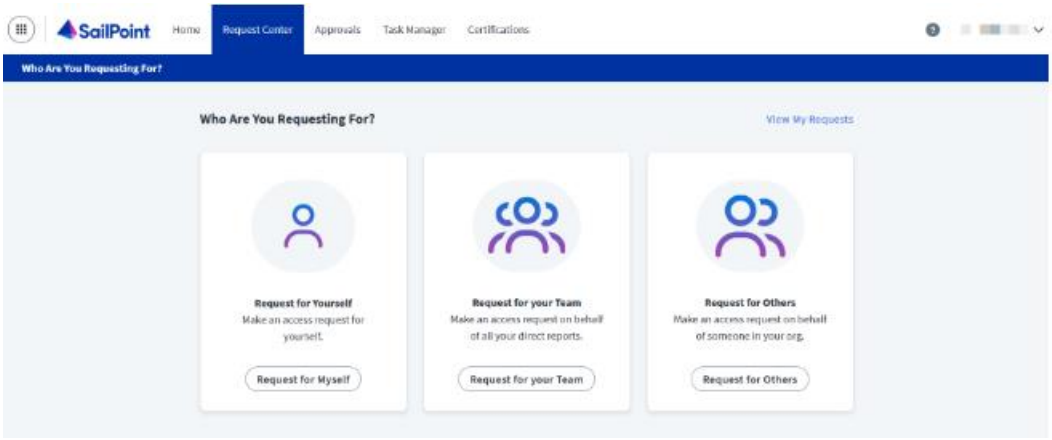

SailPoint
Home
Request Center
Approvals
Task Manager
Certifications
Search
Admin
SailPoint Trainee
Reports
All Events
Access Request Activity
Authentication Activity
Password Changes
Provisioning Activity
All Source Activity (Non-Provisioning)
Unused Cloud Management Activity
Cloud Resource Access
The Power of Identity, right at your fingertips.
With a wealth of identity and governance data at your disposal, our Dynamic Discovery Engine allows you to explore real-time data that can enhance the way you manage, evaluate, report, and secure access across your entire organization.
Not sure where to start? Need help building a query? We've got you covered.
Learn more

# Access requests

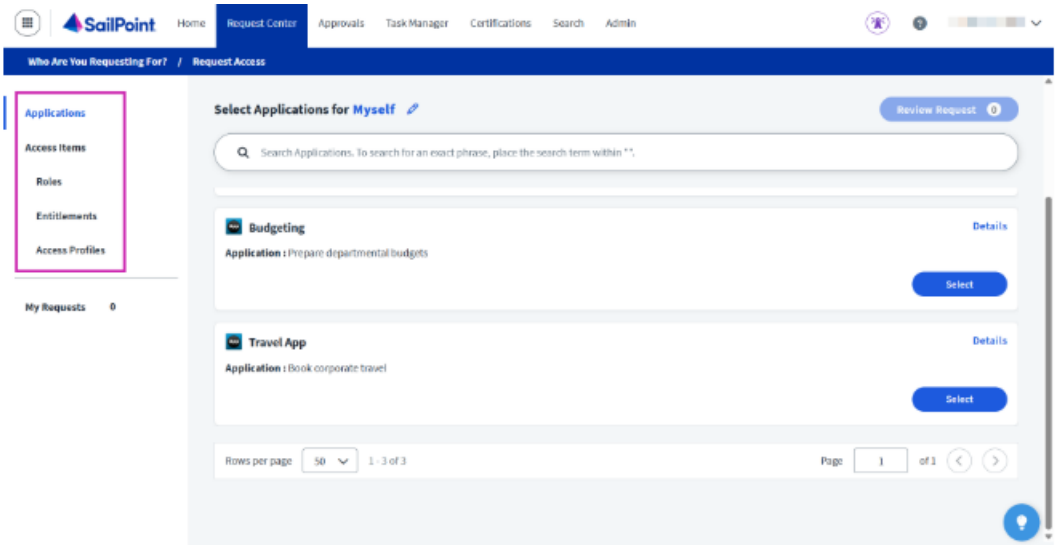
What is the access request process?

In ISC, users can easily request access to roles, entitlements, access profiles, and applications. Once submitted, the request goes through an approval process, if necessary, and once approved, access is provisioned.

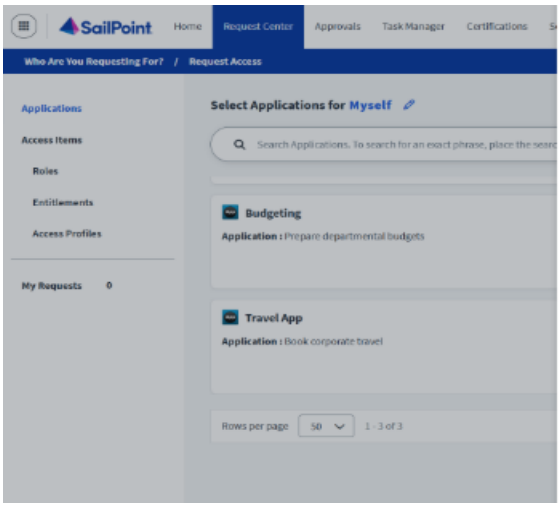
### 01 Request who should get access

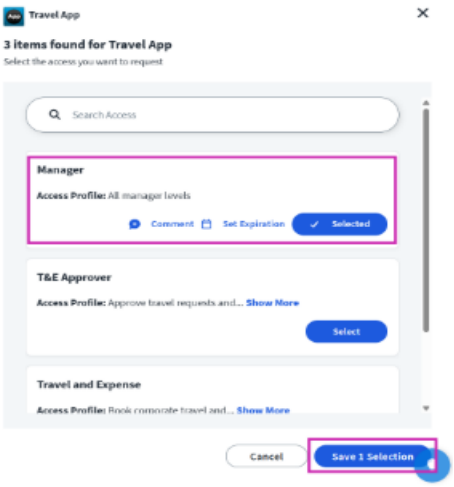



### 02 Request item

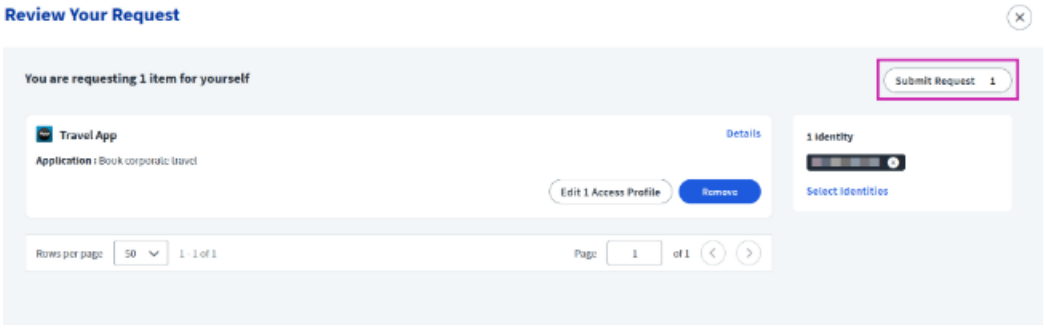


## 03 Request access profile




## 04 Review and submit request



# Tracking access requests

There are a couple ways end users can track their access requests.

Request center

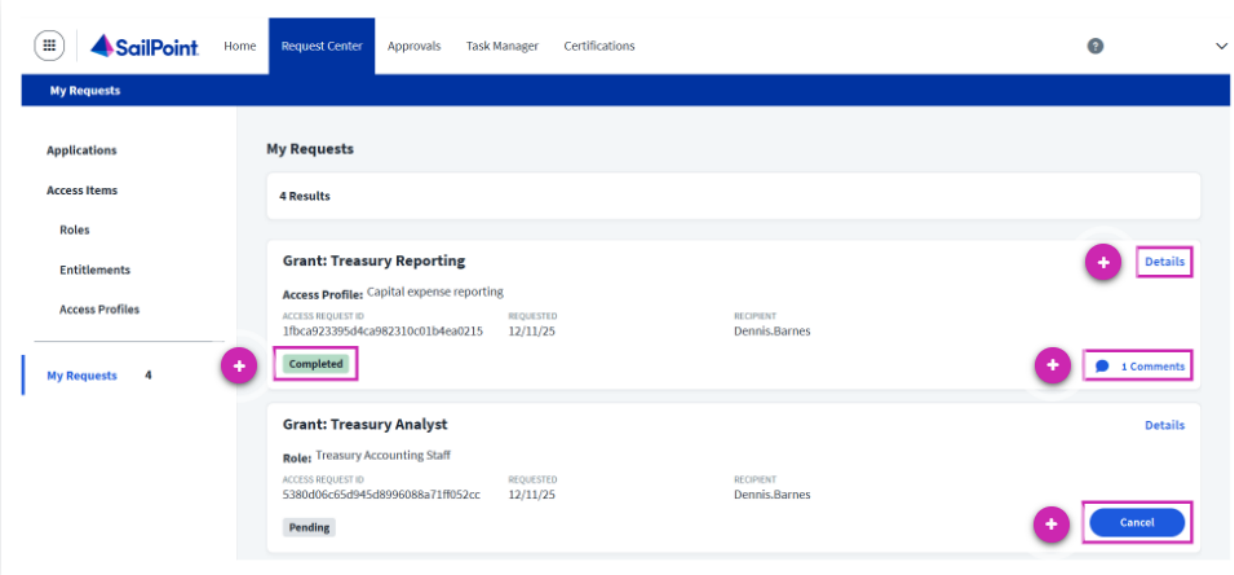
They can go to the Request Center and click View My Requests.

**Request center**

They can go to the Request Center and click View My Requests.

My Requests

This page lists all of their submitted access requests along with their current status. Users can also cancel requests, view comments, or view the details of their request



## Details panel

When you click the Details button, the Details panel will open. Here you can see the details of the process, request and account information.

**Managing requests for roles and access profiles**

You can allow users to request access to roles and access profiles by configuring them for access requests. The access items appear in the Request Center where users can submit requests, which can then be reviewed by a pre-configured set of identities to ensure that access only goes to users who need it.  The configuration process is similar for both roles and access profiles.

To allow role or access profiles requests:

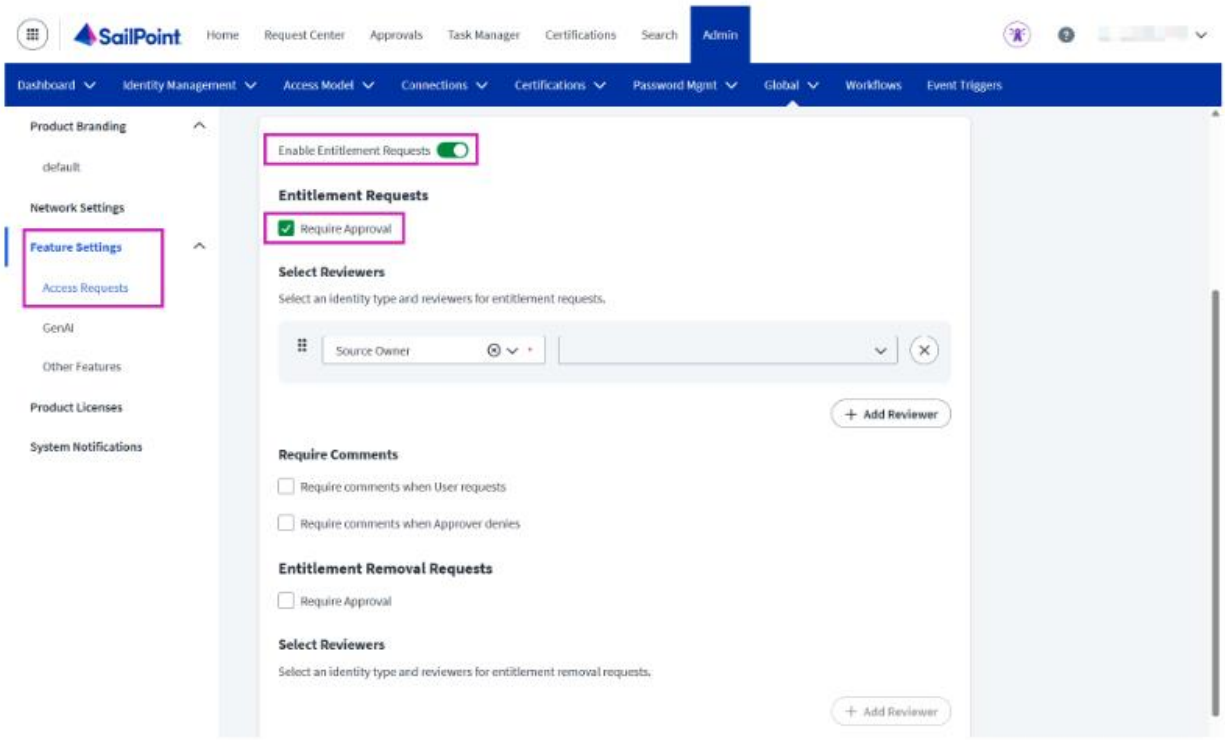
1. Navigate to Admin > Access Model > Roles or Access Profiles
2. Select Edit for the role or access profile you want to configure
3. In the Access Requests section, toggle Allow Access Requests on to enable requests for that item
4. Optionally, select Require Approval to enforce an approval process.Click on the numbered icons in the interactive image.

## Configuring access requests for entitlements

After entitlements are created and sources are uploaded, you can configure them to be requestable via the Request Center, making it easier for users to gain access to the resources they need.

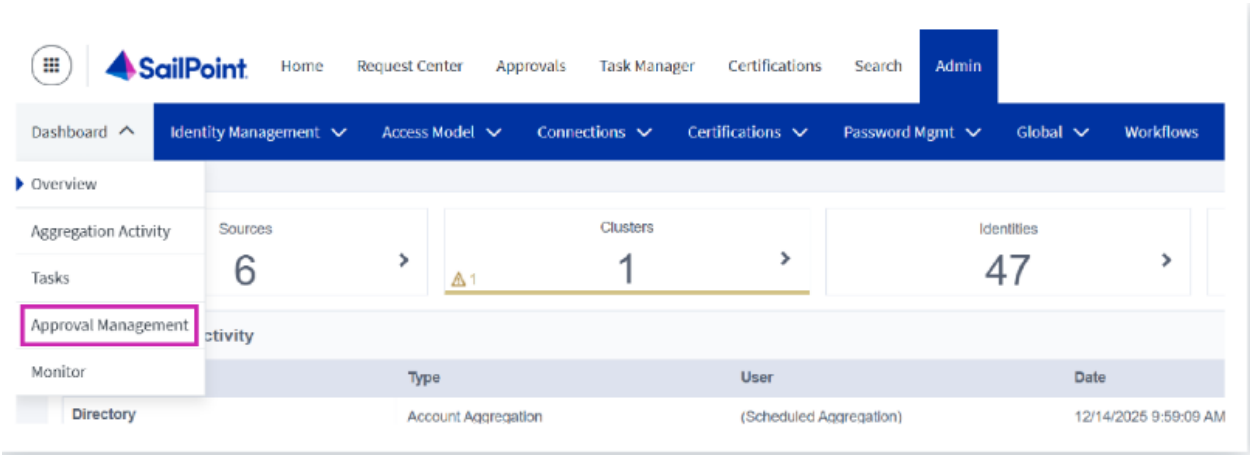
To enable entitlement requests globally:

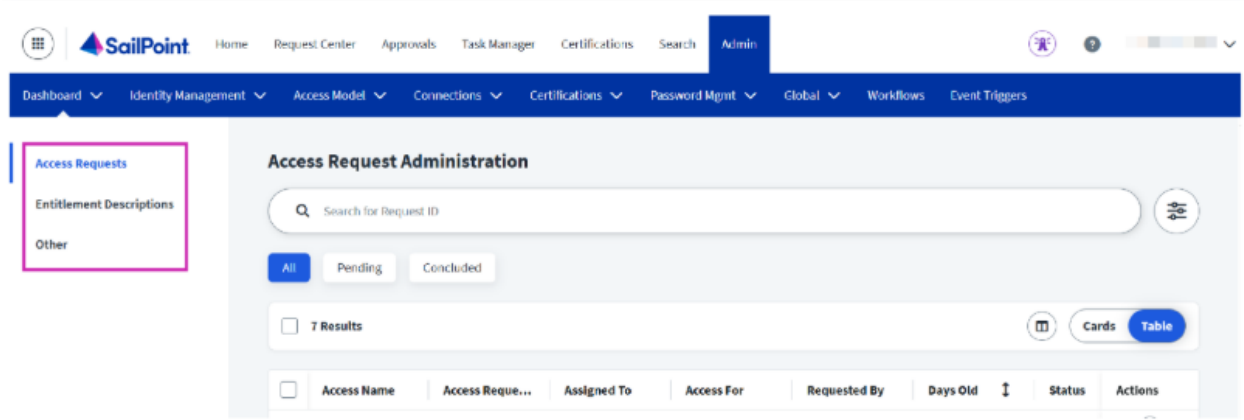
1. Go to Admin > Global > System Settings
2. Under Feature Settings in the left menu, click Access Requests
3. Check the box to enable Entitlement Requests
4. Set global approval settings for entitlement requests and removals, if needed
5. Click Save at the bottom of the page




## Access Request Administration dashboard

ISC provides a dashboard to administer many parts of access requests.  You can perform different actions such as reminding users and reassigning any pending or overdue approvals.



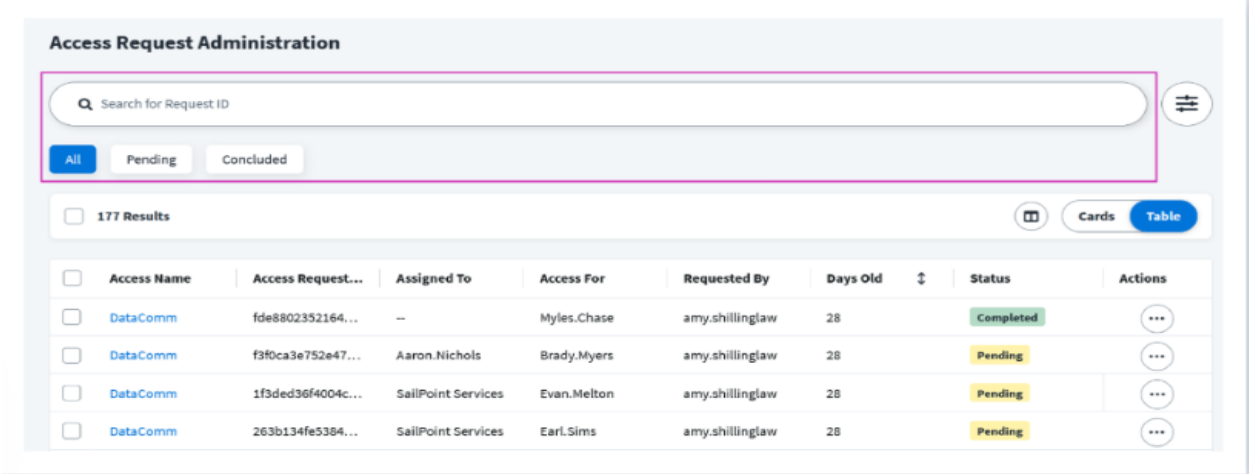



# Dashboard access

This dashboard is available to Org admins. To grant others access, you can:

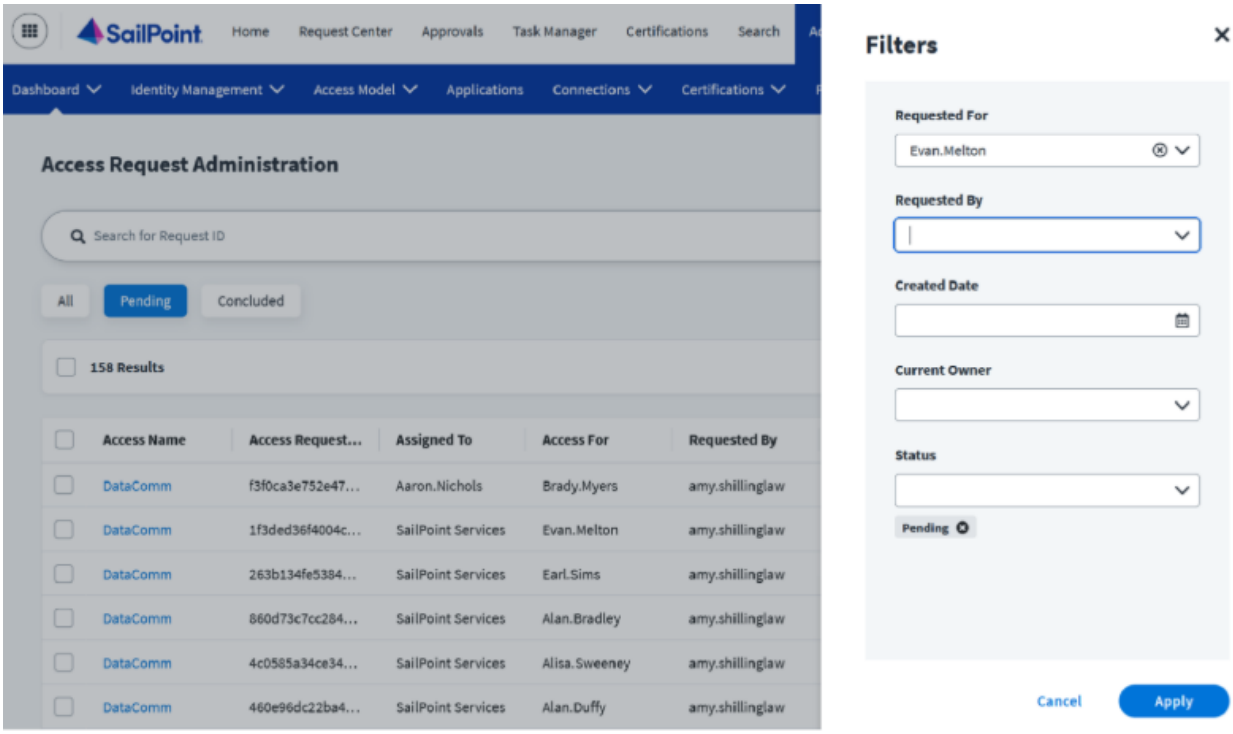
- Click **Access Request Administrator** so they can review and perform actions
- Click **Access Request Read-Only Admin** so they can only view access requests

**Searching for an access request**




**Filtering access requests**

- Requested For: Identity the access is requested for
- Requested By: Identity who submitted the request
- Created Date: Date the request was created
- Current Owner: Identity assigned to the request
- Status: Pending, Completed, Canceled, Denied, or Provisioning Failed




# What is governance?

At its core, governance is the process of managing and controlling user access to your organization's applications, data, and systems. It ensures that the right people have the appropriate access to the right resources while mitigating security risks.

## What are certification campaigns?

One of the most important governance tools in ISC is certifications. Certification campaigns are an interactive process designed to review and verify who has access to what within an organization's systems. They are a fundamental tool for managing access, reducing security risks, and ensuring compliance with audit requirements.

## Certification

A certification is an individual access review assigned to a reviewer (such as a manager or system owner). During a certification, the reviewer evaluates a user's access to enterprise systems and data and makes a decision to approve, revoke, or modify that access.

## Campaign

A set of certifications, or reviews, for all users or groups of users, entitlements, and applications.

## Campaign filter

A campaign filter is used in filter-based certification campaigns to define which users, entitlements, or applications are included in the campaign. Filters apply predefined rules to automatically determine the scope of access that will be reviewed.

# Policies

Policies help with governance by automating the detection and prevention of risky access combinations and violations of your organization's internal policies. In this module you'll learn about two types of policies, Separation of Duties (SoD) and general policies.

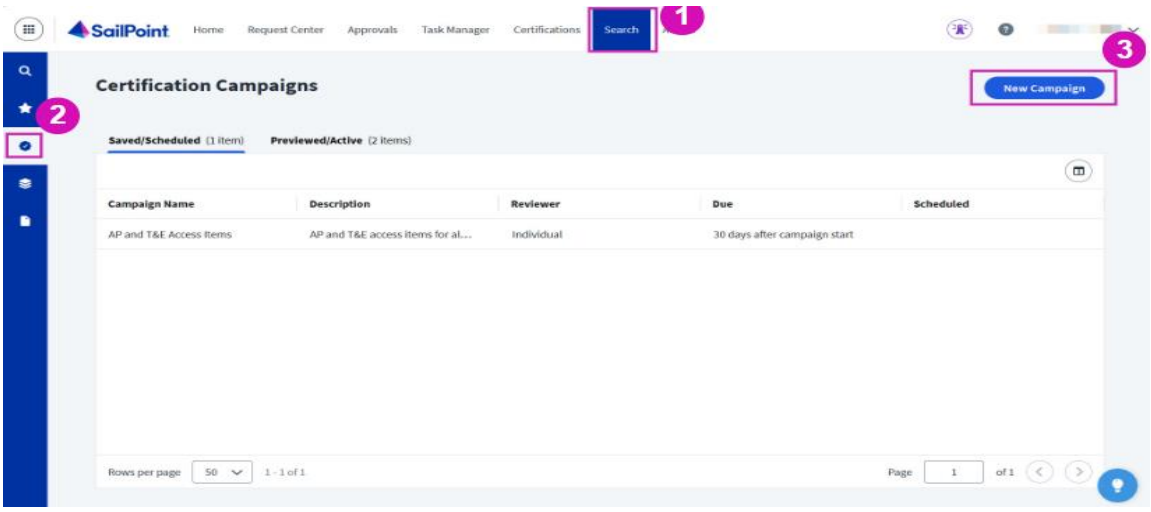
**SoD** policies help reduce risk by limiting excessive or conflicting access within critical processes. In ISC, SoD policies provide visibility into access so policy violations can be identified, assessed, and addressed.

**General policies** support this governance by helping maintain data integrity and ensuring identities align with organizational standards.

**Preparing for certification campaigns**

Follow these steps to create a certification campaign based on search query results:

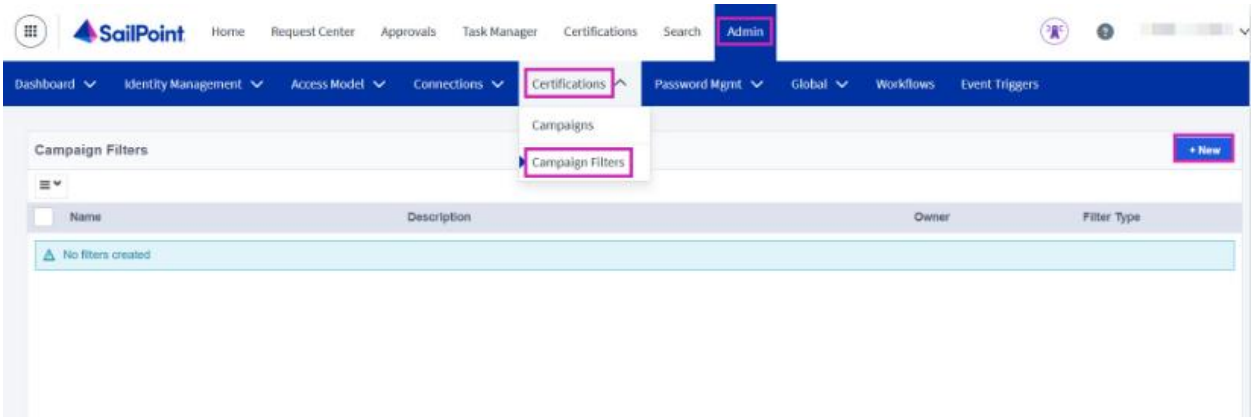
1. Select **Search** from the navigation menu
2. From the vertical Search toolbar, select the **Certification Campaigns** icon
3. Select **New Campaign** to create a new certification campaign
4. Select the tile for the type of item you'd like to certify




**Using campaign filters**

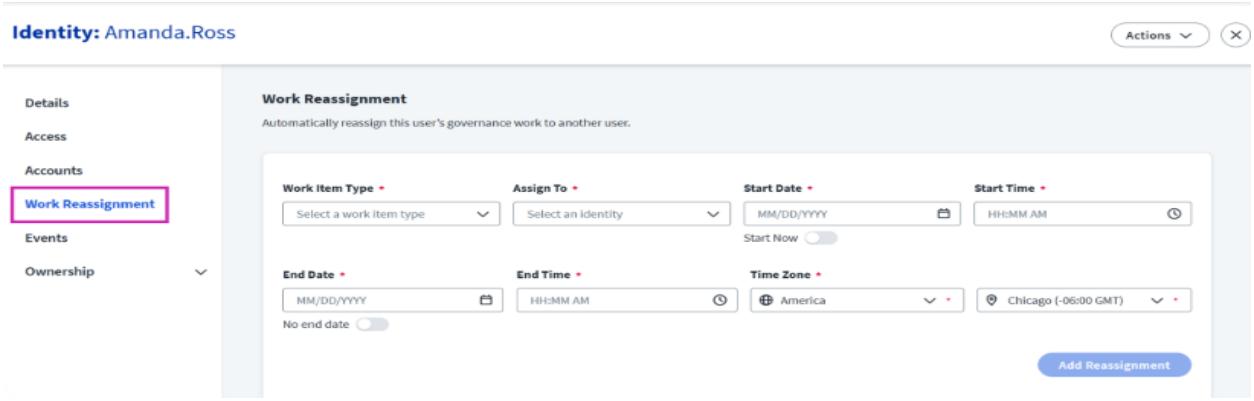
Filter-based campaigns include Manager Campaigns and Source Owner Campaigns.

Use these filters, such as department, role, or identity attributes to automatically select which users appear in the campaign.



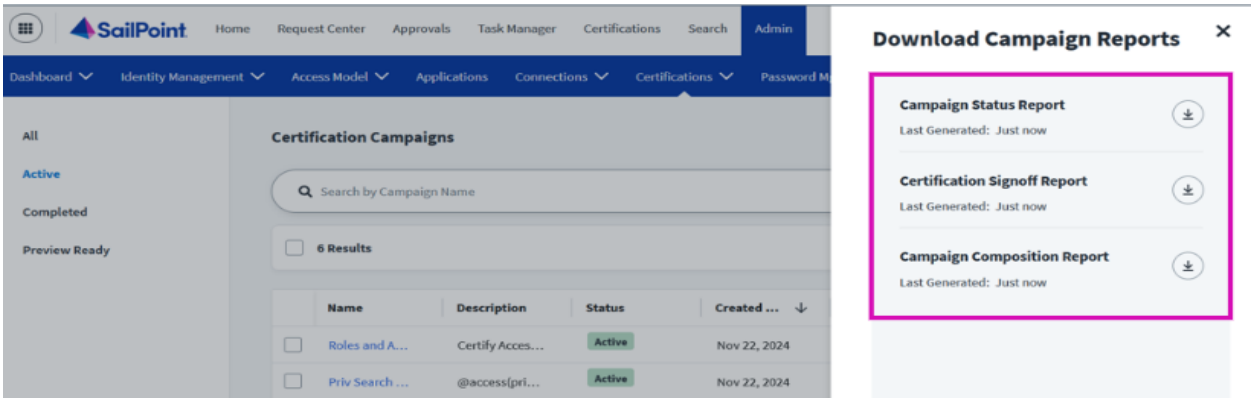

# Reassigning certifications

If you need someone other than the original reviewer to approve or revoke the access rights of users at your company, you can reassign your certifications from one user to another. This may be helpful if a reviewer is on leave, unable to finish their certifications, or if you have a team of security professionals assigned to review access.




**Monitoring campaign progress**

Admins can track the progress of certification campaigns directly from the **Admin Dashboard**.



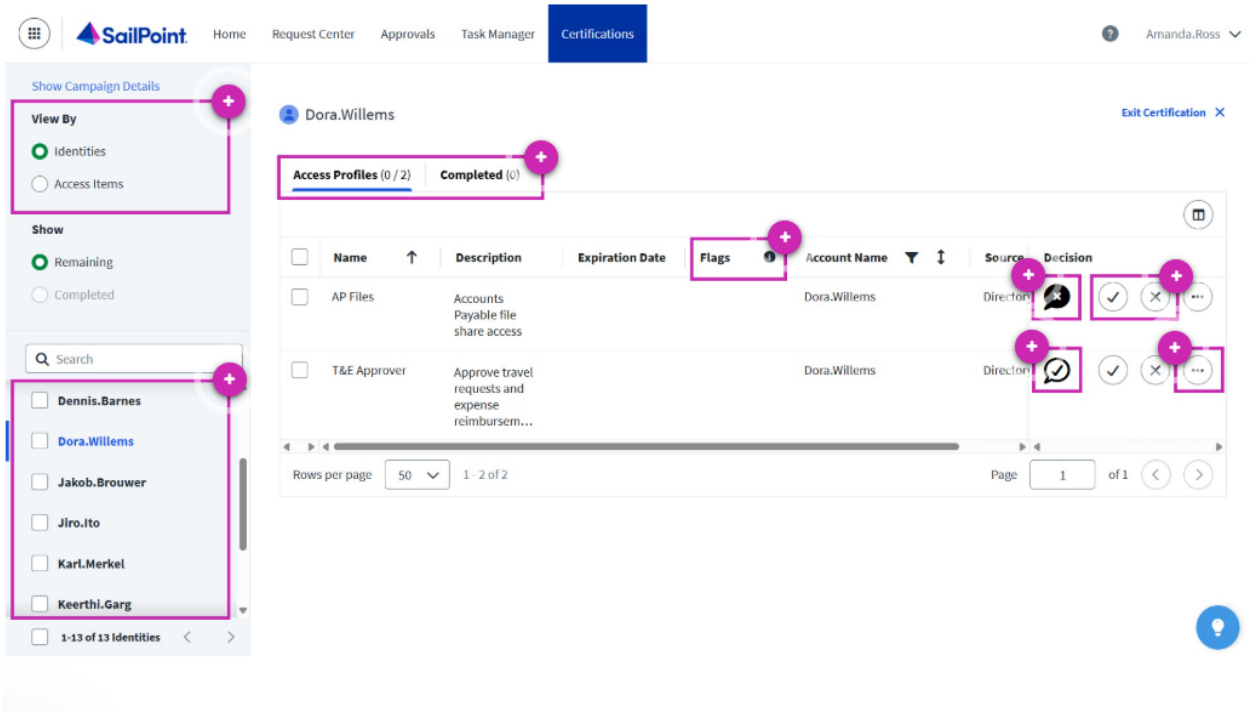
# Reviewing certifications

### Campaign details

By clicking on **DETAILS** on a certification, they'll see a summary that highlights important details about the certification campaign. They should take a moment to review this information as it provides context for what they'll be verifying.

### Review certifications

Let's look at what reviewers will see once they open a campaign



# Completing certifications

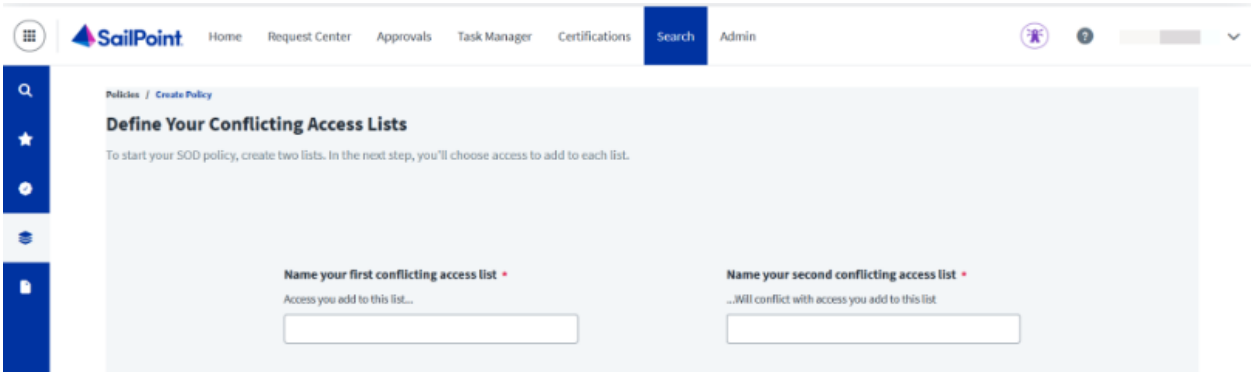
Once reviewers have reviewed each item, they're taken to a sign-off page to complete the certification.

**Policy overview**

In ISC, policies are used to identify and manage potential access risks and compliance concerns. In this lesson, you'll explore the two main policy types; SoD and general policies, and learn how policies detect and report violations, as well as how they can run automatically through subscriptions.
Policies play an important role in maintaining security and compliance. They help organizations identify risks early, meet auditing requirements, and stay aligned with internal and regulatory guidelines.

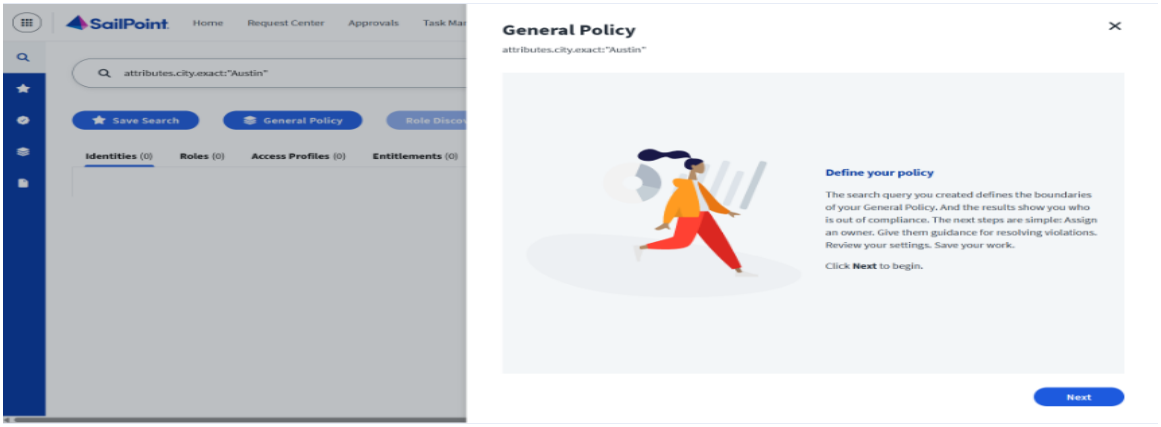
**SoD policy**

The purpose of a SoD policy is to make sure no single person has a toxic combination of access. It helps prevent situations where someone could misuse their access to commit fraud or create errors.

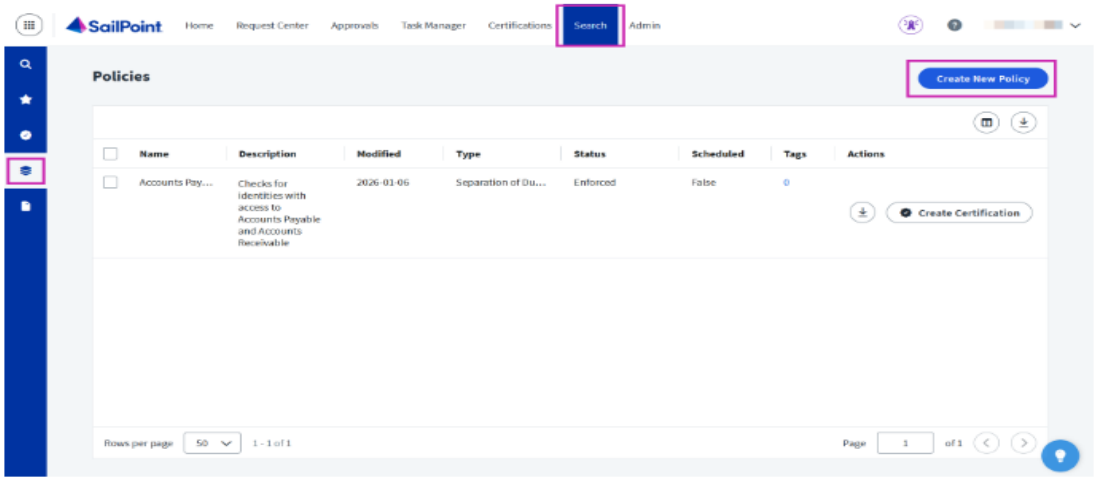


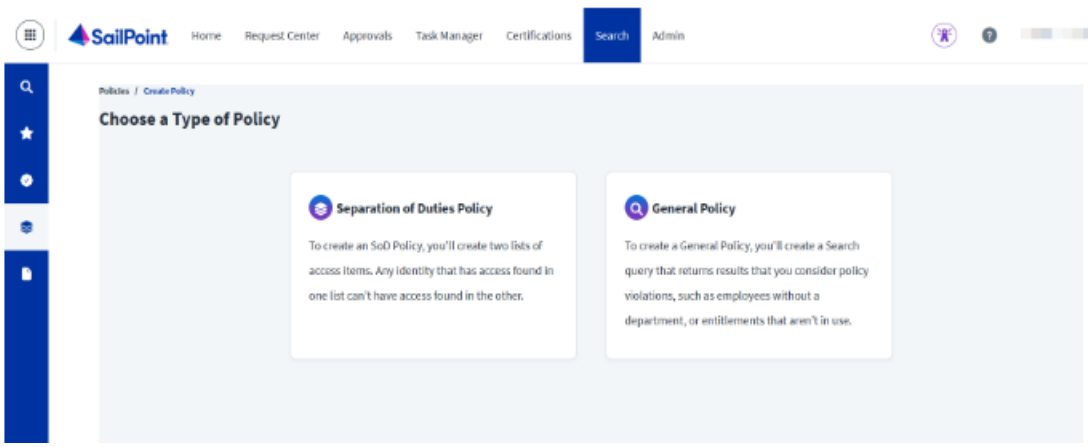
## General policy

General policies are search-based queries used to identify any unwanted data or access conditions.




## Creating policies



# Search

**Search Syntax**

Looking deeper into the organization of your data and the construction of search queries can help you create the most effective search queries.

**Search Terminology**

It's important to understand specific terminology and definitions used in SailPoint's search documentation, including:

- Terms
- First-level fields
- Second-level fields
- Arrays

**Terms**

Terms are the values that you expect to find in your data. As you saw in an earlier lesson, a term can be used on its own in a broad search to return results where the term is found anywhere in the searchable objects. Terms are also combined with fields to provide more targeted searches. Notice in all searches, the colon (:) works as equals (=).

- finance
- status:pending
- name:albert.woods

**First-Level Fields**

A first-level field is the lowest-level field, or attribute, that contains the data you're searching for.

- attributes.lastname:woods
- name:albert.woods

**Second-Level Fields**

A second-level field represents the *object* that contains the first-level field; it relates attributes together. In the search syntax, reference the object, then the attribute within it you want to find (value). Denoted with { } in the data model.

- attributes.lastname:woods

- attributes.department:sales
- @access(source.name:Mainframe)

**Arrays**

An array, a JSON structure, is nested within objects. It is a collection in the data model that contains attributes. You can query within arrays of attributes with the @ notation, specifying the object, field, and value to look for within any array.

Searching within an array will always return the top-level object in the search results.

- @access(source.name:PRISM)
- @accounts(source.name:Mainframe)

**Fields**

You can specify a field where you want to look for a value. Search will return any objects that match the search term in this field. If you specify field names, only the level you specify is searched. The more specific you get, the narrower the search. The field attribute name is case-sensitive. Examples include:

- firstName:samantha - returns all identities where the firstName field equals "samantha"
- privileged:true - returns only privileged entitlements
- email:jon.smith@acme.com

# Workflows

**What are workflows?**

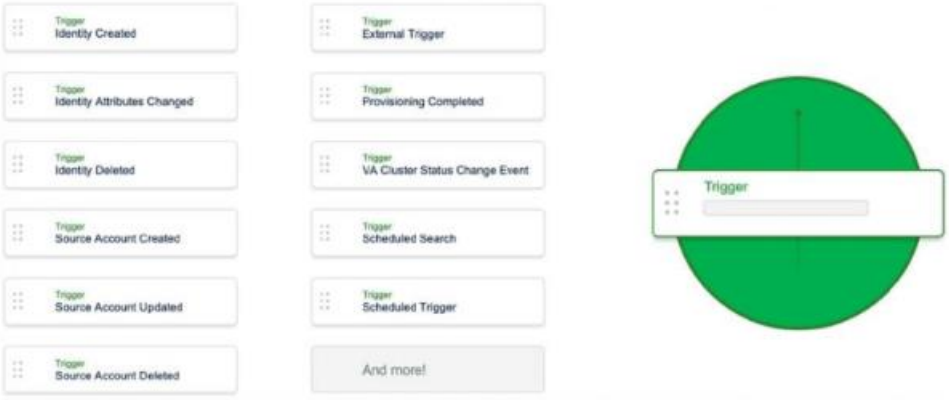
A workflow is a set of steps completed every time a specific triggering event occurs. Workflows automatically perform a series of actions within your system (that you can configure) in response to a trigger. They'll help you manage repetitive tasks through automation.

**Workflow components**

Workflows are rule-based processes with sequential steps that are composed of triggers, actions, and operators.

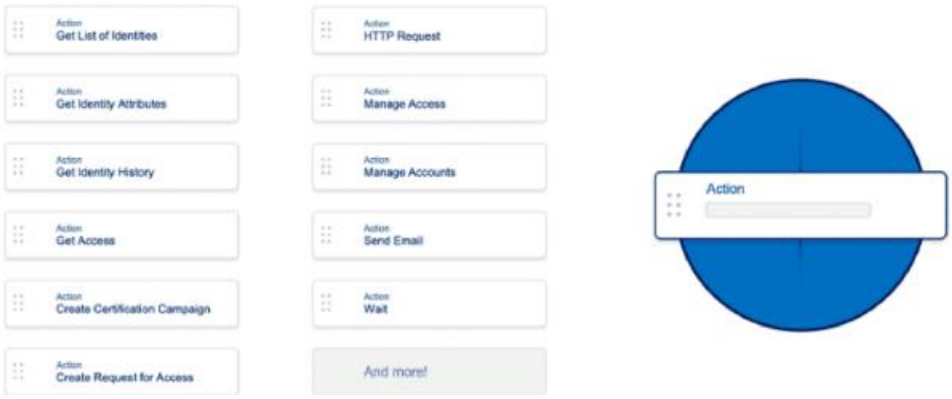
**Triggers:**

Events that initiate the workflow (such as a new identity or identity change).



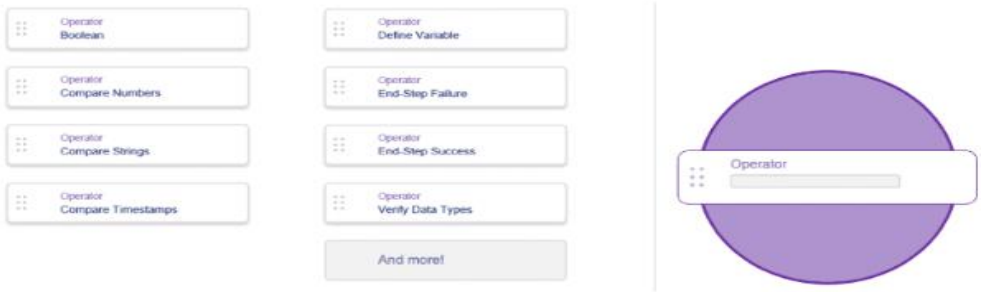

**Actions:**

Functions or operations within ISC functionality (for example, *start campaign* or *revoke access*) that are performed when the workflow runs.

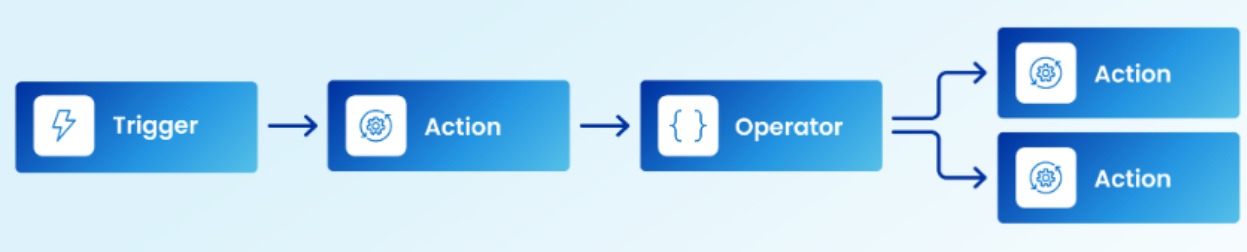



**Operators:**

A broader category of steps (such as *Is a Manager* or *Date Equals*) that act on the workflow itself by directing the data flow or making conditional choices.




**Here's a basic workflow diagram showing how the components work together:**



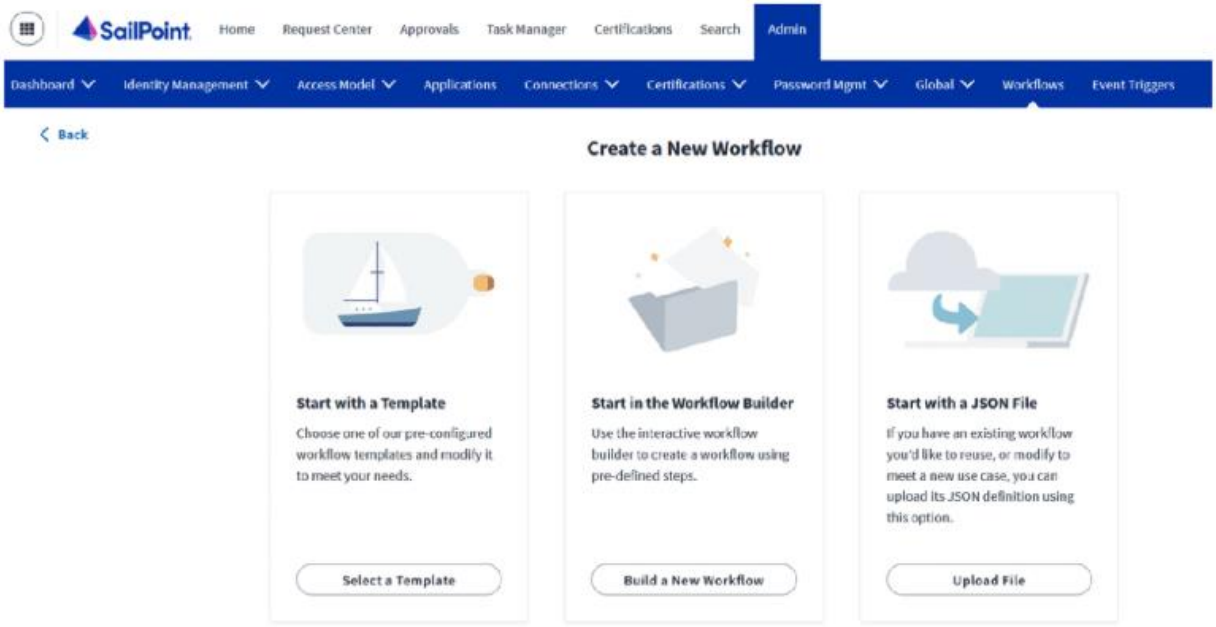

**Workflow format and syntax**

JSON

Every workflow produces a JSON script that can be downloaded, edited, shared, and uploaded. JSON is a data interchange format used to store and transmit objects and is used for things like variable definitions in workflows.

**Creating workflows**

1. Using any of the SailPoint provided templates
2. Using the UI Workflow Builder
3. Creating and uploading a JSON file




**Common workflow use cases**

Identity lifecycle changes and changes in departments are some of the most common use cases for workflows. An identity lifecycle workflow can help delegate privileged tasks by automatically provisioning or deprovisioning accesses.

1. Joiner

When new employees are onboarded, a workflow can alert a new manager, collect data from the employees using forms, and send messages about aggregation failures.

2.Mover

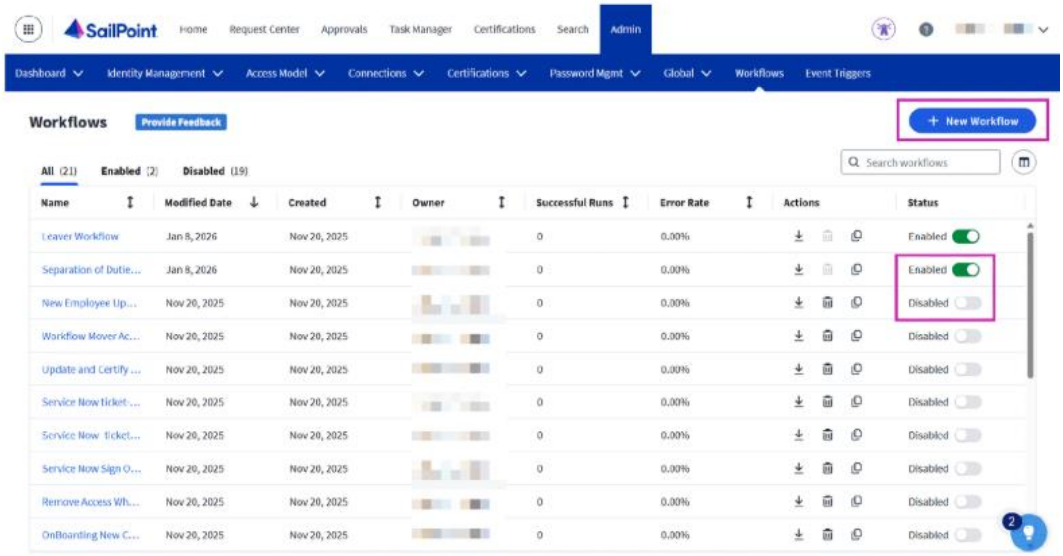
When new employees are onboarded, a workflow can alert a new manager, collect data from the employees using forms, and send messages about aggregation failures.

3.Leaver

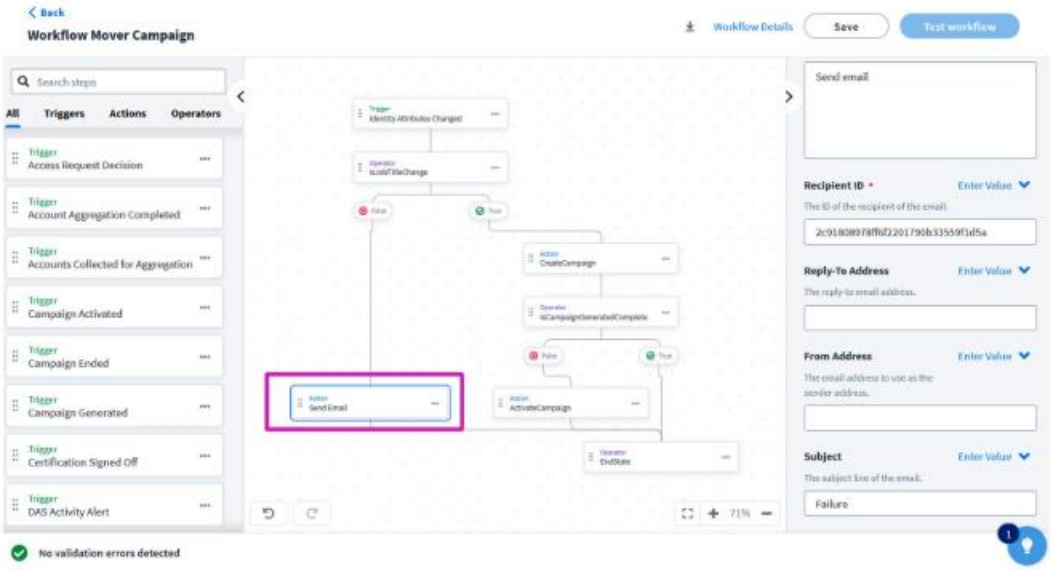
When an employee leaves an organization, a workflow can automatically generate an email and disable all associated accounts. In this case, the workflow will:

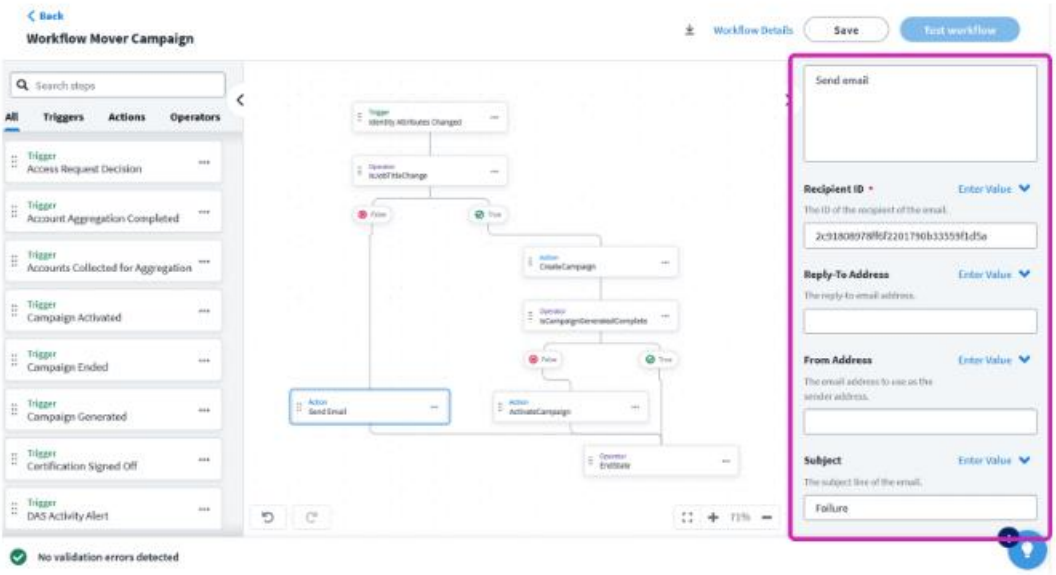
1. Trigger when an identity attribute change is detected; the trigger filter can also look for terminated identities
2. Send an email notification to the manager that an employee is leaving
3. Get accounts for the leaving identity and disable them

### Workflows landing page




### Resolving errors




Back
Workflow Mover Campaign
Workflow Details
Save
Test workflow
Search steps
All
Triggers
Actions
Operators
Trigger
Access Request Decision
Trigger
Account Aggregation Completed
Trigger
Accounts Collected for Aggregation
Trigger
Campaign Activated
Trigger
Campaign Ended
Trigger
Campaign Generated
Trigger
Certification Signed Off
Trigger
DAS Activity Alert
No validation errors detected
Send email
Recipient ID *
Enter Value
The ID of the recipient of the email.
2c91808978f8f2201790b33559f1d5a
Reply-To Address
Enter Value
The reply-to email address.
From Address
Enter Value
The email address to use as the sender address.
Subject
Enter Value
The subject line of the email.
Failure

# Transforms

**What are transforms**

Transforms are configurable objects that define easy ways to manipulate attribute data without requiring you to write code. Transforms are configurable building blocks with sets of inputs and outputs:

**How transforms work**

Transforms typically have an input(s) and output(s). The way the transformation occurs mainly depends on the type of transform.

**Multiple transform inputs**

In the previous examples, each transform had a single input. Some transforms can specify more than one input.

**Complex nested transforms**

For more complex use cases, a single transform may not be enough. It is possible to link several transforms together. Identity Security Cloud calls these 'nested' transforms because they are transform objects within other transform objects.

**Transform syntax**

Transforms are JSON objects. Prior to this, the transforms have been shown as flows of building blocks to help illustrate basic transform ideas. However at the simplest level, a transform looks like this:

```
{
  "name": "Lowercase Department",
  "type": "lower",
  "attributes": {
    "transform-attribute-1": "attribute-1-value",
    "transform-attribute-2": "attribute-2-value"
  }
}
```

**Testing transforms**

**Testing transforms in Identity Profile Mappings**

To test a transform for identity data, go to **Identities** > **Identity Profiles** and select **Mappings**. Select the transform to map one of your identity attributes, select **Save**, and preview your identity data.

**Transform best practices**

- Designing Complex Transforms - Start with small transform *building blocks* and add to them. It can be helpful to diagram out the inputs and outputs if you are using many transforms.
- JSON Editor - Because transforms are JSON objects, it is recommended that you use a good JSON editor. Atom, Sublime Text, and Microsoft Code work well because they have JSON formatting and plugins that can do JSON validation, completion, formatting, and folding. This is very useful for large complex JSON objects.
- Leverage Examples - Many implementations use similar sets of transforms, and a lot of common solutions can be found in examples.
- Same Problem, Multiple Solutions - There can be multiple ways to solve the same problem, but use the solution that makes the most sense to your implementation and is easiest to administer and understand.
- Encapsulate Repetition - If you are copying and pasting the same transforms over and over, it can be useful to make a transform a standalone transform and make other transforms reference it by using the reference type.
- Plan for Bad Data - Data will not always be perfect, so plan for data failures and try to ensure transforms still produce workable results in case data is missing, malformed, or there are incorrect values.

# Rules

In SailPoint solutions, rules serve as a flexible configuration framework implementers can leverage to preform complex or advanced configurations. Though rules allow some advanced flexibility, you must take special considerations when you are deciding to implement rules.

**Java Beanshell**

Rules in Identity Security Cloud are written in Java Beanshell, a lightweight scripting language that allows you to define custom logic and behavior within the rules. Java Beanshell provides a familiar syntax similar to Java, making it easier for implementers to create and maintain rule configurations.

There are two primary places where you can execute rules:

- **Cloud Execution** - These rules are executed in the ISC multi-tenant cloud.
- **Connector Execution** - These rules are executed on the on-premise ISC virtual appliance.

## Cloud Executed Rules

**Cloud-Executed Rules** or **Cloud Rules** typically only perform specific functions, such as calculating attribute values. Cloud Rules all execute within the SailPoint cloud and offer access to objects and data, but they do not offer any sort of external connectivity.

**Account Profile Attribute Generator**

**Overview**

This rule generates complex account attribute values during provisioning, e.g. when creating an account.

**Account Profile Attribute Generator (from Template)**

**Overview**

This rule generates complex account attribute values during provisioning, e.g. when creating an account. The rule's configuration comes from a template of values.

**Before Provisioning Rule**

**Overview**

Use this rule to modify a provisioning plan as provisioning is sent out. Do not use this rule to create new attributes. Use an account creation profile (provisioning policy) instead.

**BuildMap Rule**

**Overview**

This rule manipulates raw input data provided by the rows and columns in a file and builds a map from the incoming data. Use this rule to create a new value by combining two columns together

**Correlation Rule**

**Overview**

This rule associates or correlates an account to an identity, based on complex logic. The rule runs before configured account correlation including the default account correlation.

**Identity Attribute Rule**

**Overview**

This rule calculates and returns an identity attribute for a specific identity. This rule is known as a "complex" rule on the identity profile.

**Manager Correlation Rule**

**Overview**

This rule calculates a manager relationship between identities.

**Transform Rule**

**Overview**

This rule type is used in transforms to accomplish a specific task given configurable parameters described in the transforms attributes.

## Connector executed Rules

**Connector-Executed Rules** or **Connector Rules** are rules that are executed in the Identity Security Cloud virtual appliance, and they are usually extensions of the connector itself. The rules are commonly used to perform complex connector-related functions, so they are specific to only certain connectors.

Before Creation Rule

Executes PowerShell commands on the IQService component before a source account is created.

Before Modify Rule

Executes PowerShell commands on the IQService component before a source account is modified.

Before Delete Rule

Executes PowerShell commands on the IQService component before a source account is deleted.

After Creation Rule

Executes PowerShell commands on the IQService component after a source account is created.

After Modify Rule

Executes PowerShell commands on the IQService component after a source account is modified.

After Delete Rule

Executes PowerShell commands on the IQService component after a source account is deleted.

Build Map Rule

Calculates and transforms data from a parsed file during the aggregation process.

JDBC Build Map Rule

Calculates and transforms data from a database query result during the aggregation process. It can also perform additional calls back to the database.

JDBC Provision Rule

Executes database queries to perform provisioning of account and access for all account operations.

SAP Build Map Rule

Calculates and transforms data from SAP during the aggregation process.

SAP HR Provisioning Modify Rule

Performs SAP HR modification operations during provisioning.

Web Services Before Operation Rule

Executes before the next web-services HTTP(S) operation.

Web Services After Operation Rule

Executes after a web-services HTTP(S) operation. Often used to parse complex data.